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# Chapter 15

GELFAND-P000821

[At the upper left, crossed out: ~~22...~~]



**Uncertain reading.** The start of the crossed-out entry provisionally reads “22”. The ending is illegible.

Original source details: GELFAND-P000821

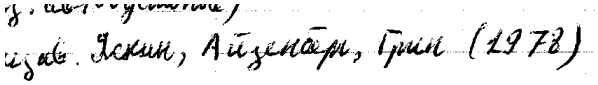
26.2.79 Gelfand Sem.

S. Gelfand from Manin’s lecture

This concerns the relation of Y.–M. solns to alg.-geom. objects

(solns need not be self-dual)

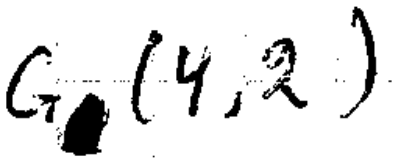
— Witten; indep. Yasskin, Isenberg, Green (1978)



**Uncertain reading.** The names are written «Яскин, Айзенберг, Грин».

Original source details: GELFAND-P000821

$\widehat{\mathbb{C}M} = G(4,2)$ (Grassmannian set of 2-dim. compl. planes in $\mathbb{C}^4$)



**Undeciphered passage.** There is a heavily inked-out entry at the lower right of $G$. Its sign is illegible.

Original source details: GELFAND-P000821

[Below $G(4,2)$:] (compl. Grassmannian set)

[Below the formula at left, a vertical line to the label:] complexif. Minkowski space

(and compactif.)

(in $G(4,2)$ there is an open subset isom. to $\mathbb{C}^4$).

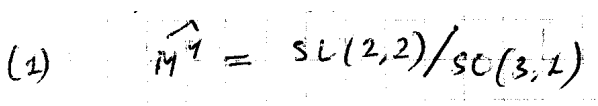
S. Gelfand

$M^4$ (Mink. space) with conf. metric is compactif. i.e. light cone

(at infinity). All conf. transf. are also $SU(2,2)$

We get the stat. group $SO(3,1)$

$$\widehat{M^4} = SU(2,2)/SO(3,1) \tag{1}$$



**Uncertain reading.** The denominator in the manuscript is $SO(3,1)$.

Original source details: GELFAND-P000821

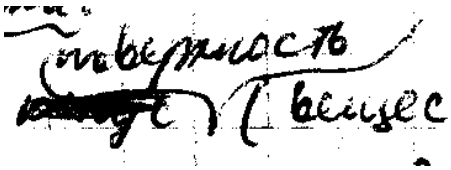
Now complexify $SU(2,2)$ and $SO(3,1)$ and take the quotient

$$(SU(2,2) = SO(4,2)) \tag{2}$$

$SU(2,2)$ — conf. group for $M^4$

Geometric picture.

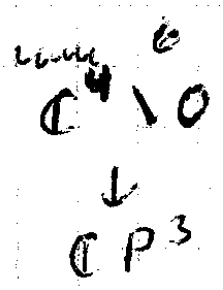
In $\mathbb{C}P^3$ consid. [crossed out: ~~cone~~] [inserted above: surface] (real 5-dimensional)



Original source details: GELFAND-P000821

**Uncertain reading.** The crossed-out word provisionally reads “cone”. “Surface” is written above it.

$$R = \{z : |z_1|^2 + |z_2|^2 - |z_3|^2 - |z_4|^2 = 0\} \subset \mathbb{C}P^3 \tag{3}$$



Original source details: GELFAND-P000821

**Diagram and typed description.**

$$(4) \quad \begin{array}{c} \text{or in } \mathbb{C}^4\backslash 0 \\ \downarrow \\ \mathbb{C}P^3 \end{array}$$

acts $GL(4,\mathbb{C})$; surf. (3) is preserved by the group $SU(2,2)$

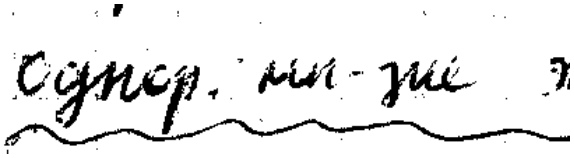
$G(4,2)$ — set of lines in $\mathbb{C}P^3$;

(or 2-dim. subspaces in $\mathbb{C}^4$)

On surf. (3) there lies wholly a set of lines of real dimension 4.

$SU(2,2)$ carries over to this set.

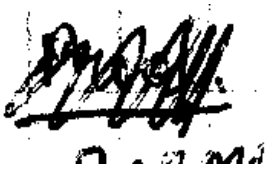
Theorem 1. Contain. set of these lines — Minkowski space.



Original source details: GELFAND-P000821

**Uncertain reading.** The first abbreviation provisionally reads «Содерж.» (“contain.”), possibly «Соотв.» (“correspond.”).

[crossed out: ~~Proof~~]



Original source details: GELFAND-P000821

**Uncertain reading.** The cancelled heading provisionally reads «Доказ.» (“Proof”).

The lines on surf. (3) are parametriz. by unit. matrices of order 2

$$(z_3, z_4) = U(z_1, z_2), \qquad U \in U(2)$$

Minkowski metric.

Consid. a fixed line $L_0$ on $R$. Consid. all other lines on $R$, that intersect it. We get a 3-dim. cone $K_{L_0}$. It is preserved under all transf. preserving $L_0$. We get $SO(3,1)$ and a cone called

the light cone.

The points of surf. $R$ can be interpreted as rays of light.

One can collect all points of intersection of $L \subset R$ with $L_0$. This gives a generator of the cone (light cone).

GELFAND-P000822

[source number: 2]

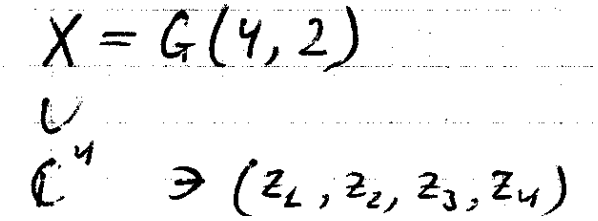
Cf.: the quaternionic version (it was discussed at some point).

Stat. subgroup:

$$GL(2,\mathbb{C}) \times \mathbb{R}^3 \qquad (?)$$

The conf. group is obtained by adding inversion $\frac{1}{\tau}$ ($\tau$ — Minkowski dist.) to the motions (the grp. of Poincaré).

$$\begin{array}{l} X = G(4,2) \\ \cup \\ \mathbb{C}^4 \ni (z_1, z_2, z_3, z_4) \end{array}$$



**Uncertain reading.** The sign below $X$ resembles $\cup$. It is set in the same vertical position.

Original source details: GELFAND-P000822

There we consid. holom. connections satisfying the Y.–M. eqs.

$G$ — a compl. Lie grp.; $\mathfrak{g}$ — its Lie algebra.

$(A_1(z), A_2(z), A_3(z), A_4(z))$ — holom. fns. with values in $\mathfrak{g}$

(holom. in an open region)

$$F_{\mu\nu}(z) = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \qquad \partial_\mu = \frac{\partial}{\partial z_\mu}. \tag{5}$$

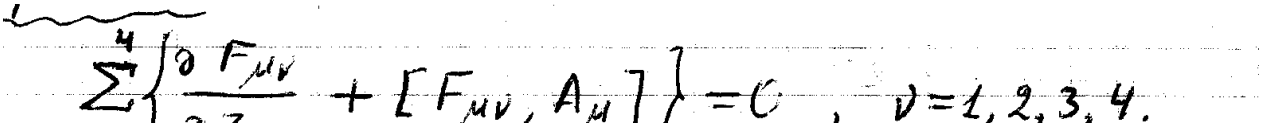
Connection form:

$$\omega = \sum A_\mu(z)\, dz_\mu$$

$F$ — the curvature form of this connection (again with values in the Lie alg.)

Y.–M. eqs.

$$\sum_{\mu=1}^{4} \left\{ \frac{\partial F_{\mu\nu}}{\partial z_\mu} + [F_{\mu\nu}, A_\mu] \right\} = 0, \qquad \nu = 1,2,3,4. \tag{6}$$



Original source details: GELFAND-P000822

**Uncertain reading.** In (6) the manuscript has $[F_{\mu\nu}, A_\mu]$ after the plus sign.

I.M.G. Landau: “the Lagrangian is a corpse” thought that one need only know the $S$-matrix, and that is all.

The corresp. real eqs. are obtained from the Lagrangian:

$$L = \operatorname{tr} \int F_{\mu\nu} F^{\mu\nu}\, d^4x \tag{7}$$

Solutions of (6) are related to certain objects of compl. analytic geometry.

From $\mathbb{C}^4$ we construct

$$\underset{\text{lines in } \mathbb{C}^4}{\mathbb{C}P^3} \qquad \text{and} \qquad \underset{\text{hyperpl. in } \mathbb{C}^4}{\mathbb{C}P^3}$$

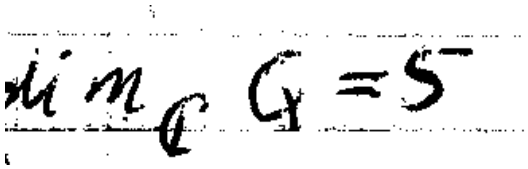
[Crossed out before the formula: ...]



**Undeciphered passage.** A separate sign is heavily crossed out. Its numeral cannot be established.

Original source details: GELFAND-P000822

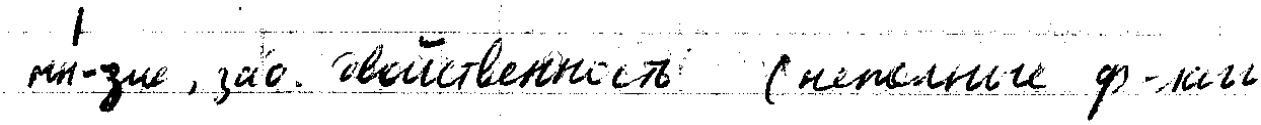
$$\mathbb{C}P^3 \times \mathbb{C}P^3 \supset Q = \{(\ell, \pi) \mid \ell \subset \pi\}, \qquad \dim_{\mathbb{C}} G = 5 \tag{8}$$



**Uncertain reading.** The letter after $\dim_{\mathbb{C}}$ provisionally reads $G$, possibly $Q$. The first $Q$ on the same line has a closed upper loop.

Original source details: GELFAND-P000822

[Vertical stroke down from $Q$:] the set defin. duality (partial fl-ags)



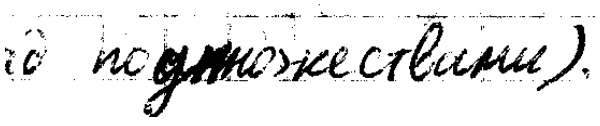
Original source details: GELFAND-P000822

**Uncertain reading.** The abbreviation before “duality” is provisionally set as «зад.» (“defin.”). The initial descender and final loop admit another reading.

2nd object.

(9) Holomorphic bundles on $Q$ with structure group $G$, which extend holom. to the 3rd infinitesim. neighborhood of $Q$ in $\mathbb{C}P^3 \times \mathbb{C}P^3$.

(In fact everything is over subsets).



**Uncertain reading.** Several letters in “subsets” are overwritten. The earlier form is illegible.

Original source details: GELFAND-P000822

All connections $\Longleftrightarrow$ all bundles (not necessarily extending to the 3rd infinit. neighborhood)

GELFAND-P000823

A. Beilinson

on $G(4,2)$ we consid. the full flag space $F$ (line, plane, hyperpl.)

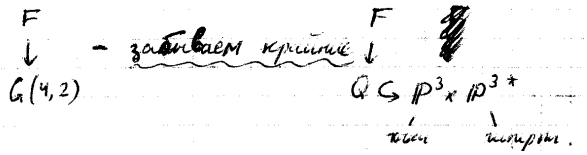






**Diagram and typed description.** 
The upper and right edges are broken. The slanted strokes have thickened parts.

Original source details: GELFAND-P000823




Original source details: GELFAND-P000823

**Diagram and typed description.**

$F$ — forget the outer ones

$G(4,2)$

$F$

$Q \hookrightarrow P^3 \times P^{3*}$

points hyperpl.

In the right diagram a hooked arrow points right from $Q$.

[Crossed out to the right of the right-hand $F$: ...]



**Undeciphered passage.** Strokes cover a vertical entry. The original sign is illegible.

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**Diagram and typed description.**



There is a separate point between the figures. A short upper-right stroke has an unclear relation to the outline.

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$(E, \nabla)$
bundle with connection
on $G(4,2)$

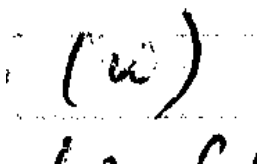
The fiber of the right bundle is isom. to $\mathbb{C}P^1$

Under the left projection it goes to $\mathbb{C}P^1$

on $\mathbb{C}P^1$ the bundle trivializes (canonically), since the curvature $= 0$ on account of the dimension being one

(fund. group $= 0$).

The connection ($\omega$) we lift to $F$ and try to descend to $Q$.



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**Uncertain reading.** The symbol in parentheses provisionally reads $\omega$. Its shape differs from $\nabla$ in the pair $(E, \nabla)$.

[Below the line] (from $G(4,2)$)

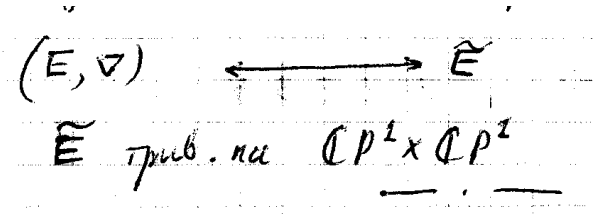
It is on $\forall$ fibers trivial $\Rightarrow$ can be carried over to $Q$

We get $\widetilde{E}$ — a holom. vect. bundle on $Q$



Original source details: GELFAND-P000823

**Uncertain reading.** The first $E$ has a thick, almost horizontal stroke above it. The wavy mark is clearer lower down.



Original source details: GELFAND-P000823

**Diagram and typed description.** $(E, \nabla) \longleftrightarrow \widetilde{E}$
$\widetilde{E}$ is triv. on $\mathbb{C}P^1 \times \mathbb{C}P^1$

Two strokes and a dot between them appear under the final line.

GELFAND-P000824

5/III-79 Gelfand Sem.

D. B. Fuchs.

On $\infty$-dimensional Lie algebras

(construction and classification).

Graded Lie alg.:

$$\mathfrak{g} = \bigoplus_{i\in\mathbb{Z}} \mathfrak{g}_i, \qquad [\mathfrak{g}_i, \mathfrak{g}_j] \subset \mathfrak{g}_{i+j}$$

hence $\mathfrak{g}_0$ — a subalg., and all the $\mathfrak{g}_i$ — $\mathfrak{g}_0$-modules.

Cartan's construction:

$$\mathfrak{g}_i = 0 \quad \text{for } i < i_0. \tag{1}$$

Transitivity: there is a homom.

$$\mathfrak{g}_i \to \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_{i-1}) \tag{2}$$

$$\mathfrak{g}_i \to \mathrm{Hom}(\mathfrak{g}_1, \mathfrak{g}_{i+1}) \tag{3}$$

$$\begin{cases} i \geq 0 \ \Rightarrow \ (2) \text{ monomorphic} \\ i \leq 0 \ \Rightarrow \ (3) \text{ monomorphic} \end{cases} \tag{4}$$

Then there is Cartan's classification thm.: there are 4 series.

1) $W_n$ — polynomial vect. fields on $\mathbb{C}^n$:

$$\sum p_i(x_1, \ldots, x_n) \frac{\partial}{\partial x_i} \tag{5}$$

deg $\frac{\partial}{\partial x_i} = -1$, [crossed out: ...] deg $x_i = 1$.



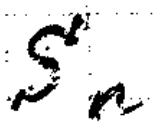
Original source details: GELFAND-P000824

**Undeciphered passage.** A sign between the two degrees is crossed out. Its original form is illegible.

Here

$\mathfrak{g}_0$ — matrix alg.(!) (gen. by the fields $x_j \frac{\partial}{\partial x_i}$).

2) divergence-free vect. fields $S_n$



Original source details: GELFAND-P000824

**Uncertain reading.** A short rising stroke joins the upper-right edge of $S$. It may be the end of the letter or a prime mark.

3) Hamiltonian $H_n$

4) contact $K_n$ pres. the cont. struct. in $\mathbb{C}^{2n+1}$

(description of the grading is omitted)

(all are polynomial)

Idea of the proof:

Cart. prolongation.

For simplicity assume $i_0 = -1$ (in (1))

$$\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \cdots \tag{6}$$

From $\mathfrak{g}_0$ and $\mathfrak{g}_{-1}$ everything is recovered

GELFAND-P000825

[source number: 10]

We have:

$$\mathfrak{g}_1 \subset \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_0) \tag{7}$$

$$f \in \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_0), \quad \alpha_1, \alpha_2 \in \mathfrak{g}_{-1} \quad \Rightarrow$$

$$\Rightarrow$$

$$f(\alpha_1)\alpha_2 = f(\alpha_2)\alpha_1 \tag{8}$$

(from the Jacobi id.)



**Uncertain reading.** The subscript on the second $\alpha$ on the left is retraced. Its visible form reads 2.

Original source details: GELFAND-P000825

Let

$$\mathfrak{g}_1 = \{f \in \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_0) \mid f(\alpha_1)\alpha_2 = f(\alpha_2)\alpha_1\} \tag{9}$$

$$\mathfrak{g}_2 \subset \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_1) \subset \mathrm{Hom}(\mathfrak{g}_{-1}, \mathrm{Hom}(\mathfrak{g}_{-1}, \mathfrak{g}_0)) = \\ = \mathrm{Hom}(\mathfrak{g}_{-1} \otimes \mathfrak{g}_{-1}, \mathfrak{g}_0) \tag{10}$$

Jacobi id. $\Rightarrow$

$$f(\alpha_1 \otimes \alpha_2)\alpha_3 \text{ — symm. in all 3 vars.} \tag{11}$$

Put

$$\mathfrak{g}_2 = \left\{ \begin{matrix} f \in \mathrm{Hom}(\mathfrak{g}_{-1} \otimes \mathfrak{g}_{-1}, \mathfrak{g}_0) \mid \\ f(\alpha_1 \otimes \alpha_2)\alpha_3 \text{ symm. in all vars.} \end{matrix} \right\} \tag{12}$$

and then

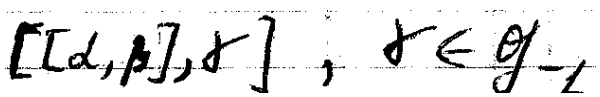
$$\mathfrak{g}_k = \left\{ \begin{matrix} f \in \mathrm{Hom}(\bigotimes^k \mathfrak{g}_{-1}, \mathfrak{g}_0) \mid \\ f(\alpha_1 \otimes \cdots \otimes \alpha_k)\alpha_{k+1} \text{ is symmetric} \end{matrix} \right\} \tag{13}$$

The commutator in $[\bigoplus \mathfrak{g}_k]$ — ?

with $\mathfrak{g}_{-1}$ the commut. is already defined, with $\mathfrak{g}_0$ — likewise.

$$\alpha \in \mathfrak{g}_k, \quad \beta \in \mathfrak{g}_\ell \quad \Rightarrow \quad \text{to def. } [\alpha, \beta] \text{ one needs to specify}$$

$$[[\alpha, \beta], \delta], \qquad \delta \in \mathfrak{g}_{-1}$$

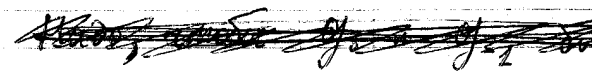


**Uncertain reading.** The same letter in both places provisionally reads $\delta$. $\gamma$ is also possible.

Original source details: GELFAND-P000825

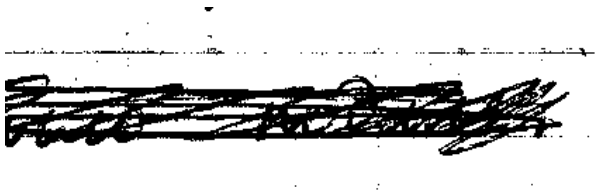
Jacobi id. $\rightsquigarrow$ into terms def. by induction.

[Crossed out: ~~We need $\mathfrak{g}_0$ and $\mathfrak{g}_{-1}$ to be ...~~]



Original source details: GELFAND-P000825

**Uncertain reading.** The start of the crossed-out line is provisional. Strokes cross the connector between the two $\mathfrak{g}$ terms and the sign before the index 1.



Original source details: GELFAND-P000825

**Undeciphered passage.** The end of the crossed-out line is illegible.

$\mathfrak{g}_0, \mathfrak{g}_{-1}$ can be arbitrary. Simplicity is an addl. condition.

But as a rule we get a finite-dimensional algebra!!

$\infty$-dimensionality — luck.

Another constr.

Let $\mathfrak{g}_{-1}, \mathfrak{g}_0$ be given, $\quad \mathfrak{g}_0 \subset \mathrm{End}(\mathfrak{g}_{-1})$.

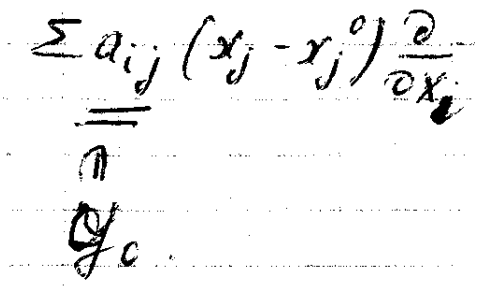
GELFAND-P000826

In $\mathfrak{g}_{-1}$ we consid. poly. vect. fields. A subalgebra is specified in it.

$$\sum_i p_i \frac{\partial}{\partial x_i} \in \mathfrak{g}, \quad \text{if } \forall x^0 \in \mathfrak{g}_{-1},$$

the lin. part of the Tayl. expan. at pt. $x^0$ belongs to $\mathfrak{g}_0$.

$$\sum a_i \frac{\partial}{\partial x_i} + \sum \underset{\underset{\mathfrak{g}_0}{\cap}}{\underline{\underline{a_{ij}}}} (x_j - x_j^0) \frac{\partial}{\partial x_i} \tag{14}$$



Original source details: GELFAND-P000826

**Uncertain reading.** Below the doubly underlined $a_{ij}$ the manuscript has a rotated membership sign, then $\mathfrak{g}_0$.

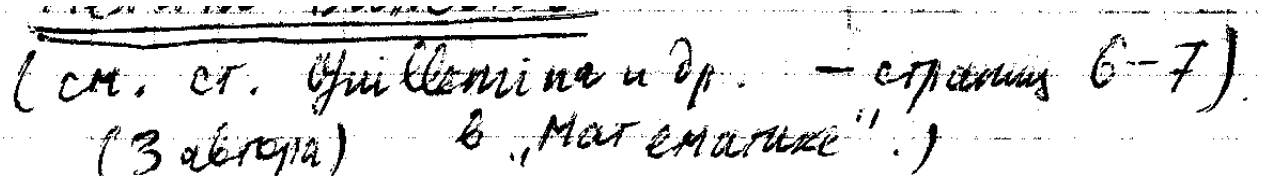
Natural grading — by degrees.

This coincides with the Lie alg. constructed above

Wilson's lemma

(see the paper by Guillemina et al. — pages 6–7).

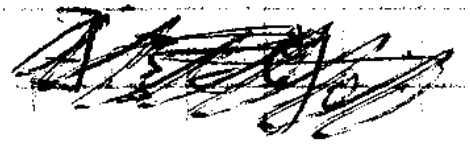
(3 authors) in "Matematika".)



Original source details: GELFAND-P000826

**Uncertain reading.** The name provisionally reads "Guillemina". The final «a» may be a Russian case ending. The title on the lower line is "Matematika".

Cart. prolong. leads to an $\infty$-dim. Lie alg. $\Longleftrightarrow$ [crossed out: $\exists \xi \in \mathfrak{g}_0$]



Original source details: GELFAND-P000826

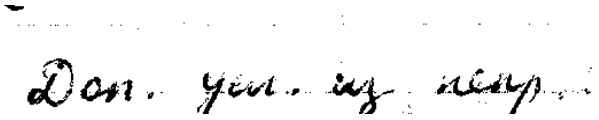
**Uncertain reading.** The crossed-out formula provisionally reads $\exists \xi \in \mathfrak{g}_0$. Several strokes overlap.

$$\exists \xi \underset{\mathfrak{g}_0}{\cap} : \quad \xi : \mathfrak{g}_{-1} \to \mathfrak{g}_{-1} \quad \text{has rank } 1 \tag{15}$$

(this has nothing to do with simplicity)

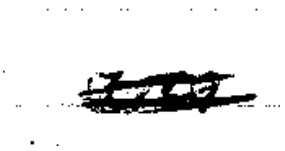
Addl. cond. from the lit. [crossed out: ...] something must be reductive,

—"———"— irreducible



Original source details: GELFAND-P000826

**Uncertain reading.** The abbreviation after "from" is provisionally set as «литр.» ("lit.").

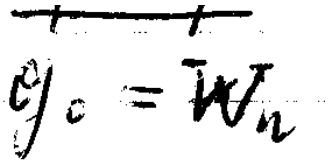


Original source details: GELFAND-P000826

**Undeciphered passage.** A short word is crossed out. Its original letters are illegible.

Example.

$\mathfrak{g}_0 = W_n, \qquad \mathfrak{g}_{-1} = \text{polynomials} = \mathbb{C}[x_1, \ldots, x_n]$



Original source details: GELFAND-P000826

**Uncertain reading.** There is a short separate stroke above the left edge of $W$. It may be a serif or a bar above the letter.

Cart. expans. (1st method) $\Rightarrow$ an alg. barely different from Ham. fields

(alg. of Poisson — there is a certain bracket)

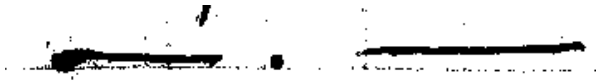


**Diagram and typed description.** ___ . ___

Original source details: GELFAND-P000826

Another method, leads to the algebras Kac-Moody [«Каца-» is written above “Kac”]

($\Rightarrow$ Kac–Macdonald id.)

A sq. (Cartan) matrix is needed

$$A = (n_{\alpha\beta}) \quad \text{of ord. } N, \quad \text{integer entries} \tag{16}$$

$$\begin{cases} n_{\alpha\alpha} = 2, \quad \text{off diag. — integers } \leq 0, \\ n_{\alpha\beta} \text{ and } n_{\beta\alpha} \text{ become } 0 \text{ only simultaneously.} \end{cases} \tag{17}$$

(18) Symmetrizability: bring to a symm. matrix in the ambient space by dilations along the coord. axes:

$$n_{\alpha\beta} \mapsto n_{\alpha\beta} \frac{\lambda_\alpha}{\lambda_\beta}$$

GELFAND-P000827

[source number: 11]

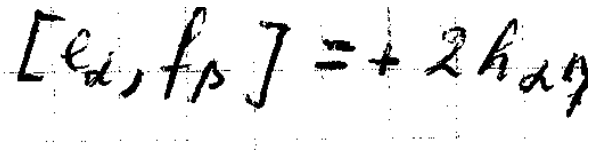
($\Rightarrow \exists$ inv. form)

A Lie alg. is def. by gen. and rels.

(19) Generators: $f_\alpha, h_\alpha, e_\alpha, \quad \alpha = 1, \ldots, N$

Relations:

$$\begin{cases} [h_\alpha, h_\beta] = 0, \quad [h_\beta, e_\alpha] = n_{\alpha\beta} e_\alpha, \quad [h_\beta, f_\alpha] = -n_{\alpha\beta} f_\alpha, \\ [e_\alpha, f_\beta] = 0 \text{ for } \alpha \neq \beta, \quad [e_\alpha, f_\beta] = +2h_\alpha \text{ for } \alpha = \beta, \end{cases} \tag{20}$$

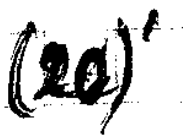


**Uncertain reading.** The final relation in (20) reads $+2h_\alpha$.

Original source details: GELFAND-P000827

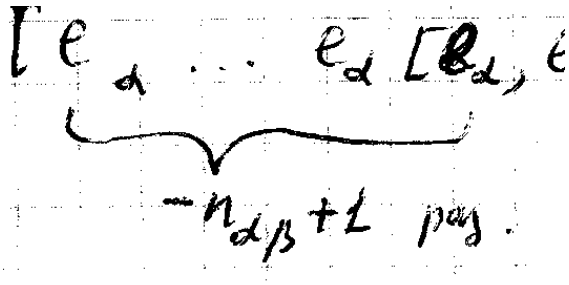
$$[\underbrace{e_\alpha \ldots e_\alpha [e_\alpha}_{-n_{\alpha\beta}+1 \text{ times}}, e_\beta] \ldots] = 0 \tag{20}'$$

[At right:] and similar conds. for $f$



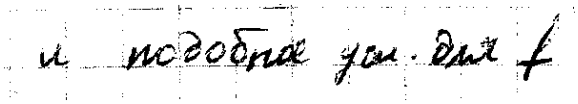
**Uncertain reading.** The number is retraced and reads $(20)'$. The preceding strokes are indistinct.

Original source details: GELFAND-P000827



Original source details: GELFAND-P000827

**Uncertain reading.** The innermost $e_\alpha$ is retraced. Below the brace is "$-n_{\alpha\beta}+1$ times".



Original source details: GELFAND-P000827

**Uncertain reading.** Before "for" the abbreviation provisionally reads «усл.» ("conds.").

$f_\alpha, h_\alpha, e_\alpha$ are lin. indep..

There are 2 gradings.

Fine

$$\Gamma = \mathbb{Z}\Big[\underset{\text{symbol}}{\alpha}\Big] \quad \text{— lattice gen. by the symbols } \alpha.$$

$$\begin{cases} \mathfrak{g} = \bigoplus\limits_{\gamma\in\Gamma} \mathfrak{g}_\gamma \\ f_\alpha, h_\alpha, e_\alpha \quad \text{homog. of degrees} \\ -\alpha, 0, \alpha \end{cases} \tag{21}$$

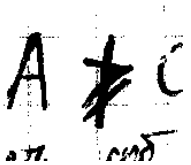
and the grading then extends uniquely.

$$\begin{cases} \underline{\text{Coarse}}: \quad \mathfrak{g} = \bigoplus\limits_{i\in\mathbb{Z}} \mathfrak{g}_i \\ \begin{array}{ccc} f_\alpha & h_\alpha & e_\alpha \\ -1 & 0 & 1 \end{array} \end{cases} \tag{22}$$

$$\underset{(\text{eigenv.} > 0)}{A} \gg 0 \quad \Rightarrow \quad \text{the alg. is finite-dimensional (!)} \tag{23}$$

$$\underset{\substack{(\text{there is an eigenv.} = 0,\\ \text{all} \geq 0)}}{A} \geq 0 \quad \Rightarrow \quad \begin{array}{c}\text{the alg. is } \infty\text{-dimensional, but growth 1} \\ (\dim \mathfrak{g}_i \text{ are bounded collectively})\end{array} \tag{24}$$

$$\underset{(\text{there is an eigenvalue} < 0)}{A} \not\geq 0 \quad \Rightarrow \quad \text{has exp. growth} \tag{25}$$



Original source details: GELFAND-P000827

**Uncertain reading.** A descending stroke crosses the inequality sign. It is provisionally set as $\not\geq$. A retraced $\neq$ is also possible.

For finite-dimensional algebras, Cartan matrs. also arise.

$$\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & & 0 \\ & & \ddots & -1 \\ 0 & & -1 & 2 \end{pmatrix} \quad - \quad sl(n+1)$$

GELFAND-P000828

Roots of Kac–Moody algs.:

$$\gamma \neq 0, \quad \mathfrak{g}_\gamma \neq 0 \quad \Rightarrow \quad \gamma \text{ — a root}$$

Roots, e.g., the $\alpha$ — simple pos. roots

$$\text{pos. roots} — \sum z_\alpha \alpha, \quad z_\alpha \geq 0$$

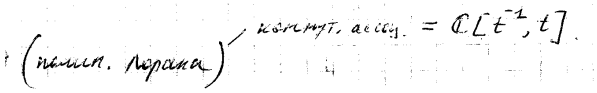
Most interesting.
degenerate $A \geq 0$ —
— then there is a center.

$$\begin{aligned} &\sum k_\beta n_{\alpha\beta} = 0 \quad \Rightarrow \\ &\Rightarrow \quad \sum k_\beta h_\beta \in \text{ the center} \end{aligned} \tag{26}$$

Example. Matrs. of ord. 2.

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} \tag{27}$$

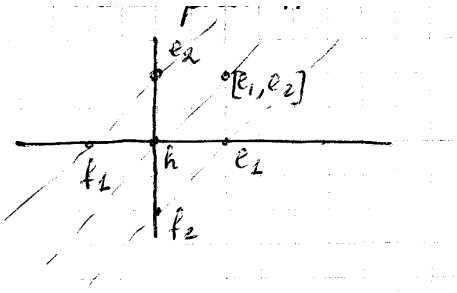
$$\begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix} \quad \text{and its symmetric counterpart.} \tag{28}$$




Original source details: GELFAND-P000828

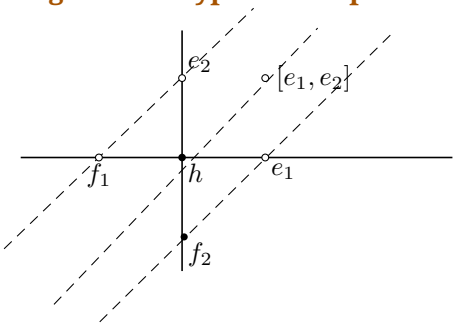
**Diagram and typed description.**
commut., assoc. $= \mathbb{C}[t^{-1}, t]$.
(27) $\Rightarrow sl(2) \otimes$ (Laurent polyn.)

$\Gamma$ — a lattice in the plane.




Original source details: GELFAND-P000828

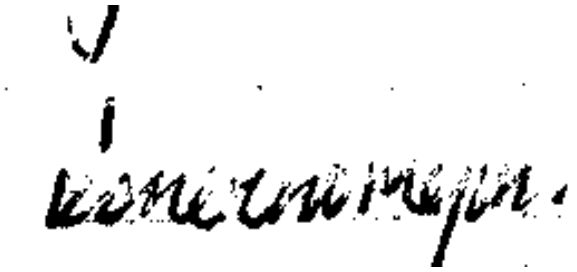
**Diagram and typed description.**




The broken diagonals are drawn without arrowheads. The point $f_2$ merges with the vertical line.

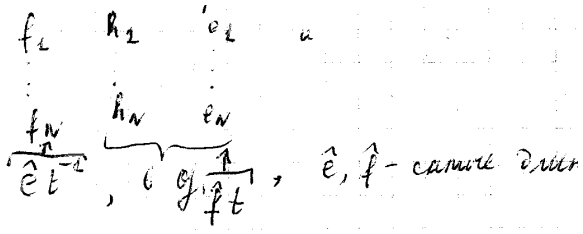
$e, h, f$ multipl. in all poss. powers.

In general



Original source details: GELFAND-P000828

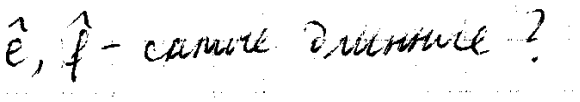
**Diagram and typed description.**
$\mathfrak{g} \otimes \mathbb{C}[t^{-1}, t]$ — a Kac–Moody alg..
|
finite-dimension.



Original source details: GELFAND-P000828

**Diagram and typed description.**
$f_1 \quad h_1 \quad e_1$ and
$\vdots \quad \vdots \quad \vdots$
$f_N \quad h_N \quad e_N$
$\uparrow \quad \uparrow \quad \uparrow$
$\hat{e}t^{-1}$ in $\mathfrak{g}$ $\hat{f}t$

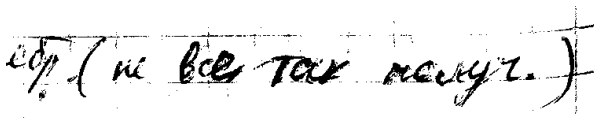
$\hat{e}, \hat{f}$ — the longest ?



Original source details: GELFAND-P000828

**Uncertain reading.** The manuscript has a question mark immediately after “longest”.

There are also other ways of constr. these algebras (not all are obtained this way)



Original source details: GELFAND-P000828

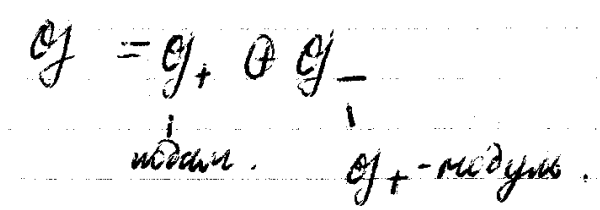
**Uncertain reading.** The opening words in parentheses read «не все» (“not all”). «все так» is retraced.

GELFAND-P000829

[source number: 12]

2nd constr. Suppose

There is an involution in $\mathfrak{g}$: $\tau : \mathfrak{g} \to \mathfrak{g}$. Then



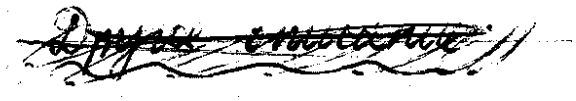
Original source details: GELFAND-P000829

**Diagram and typed description.**
$\mathfrak{g} = \mathfrak{g}_+ \oplus \mathfrak{g}_-$
|  \
subalg. $\mathfrak{g}_+$-module.

The algebra is a subalgebra of $\mathfrak{g} \otimes \mathbb{C}[t, t^{-1}]$
of the form

$$\ldots \oplus \mathfrak{g}_- t^{-1} \oplus \mathfrak{g}_+ \oplus \mathfrak{g}_- t \oplus \mathfrak{g}_+ t^2 \oplus \ldots$$

$\mathfrak{g} = sl(3)$, $\tau$ : refl. in the diag. (−transpose)
[Crossed out: ~~Other algebras~~]



Original source details: GELFAND-P000829

**Uncertain reading.** The crossed-out heading provisionally reads “Other algebras”. A wavy line is drawn underneath.

[At left, crossed out: $\mathfrak{g}$]



Original source details: GELFAND-P000829

**Uncertain reading.** A lower loop is visible under the strokes. The sign resembles $\mathfrak{g}$.

Other Lie algebras.
gen. $\varepsilon_i,\ i \in \mathbb{Z}, \quad [\varepsilon_i, \varepsilon_j] = a_{ij}\varepsilon_{i+j}$
there are 3 simple ones
1st:

$$a_{ij} = \begin{cases} 1, 0, -1 \\ \equiv j - i \mod 3 \end{cases}$$

These 2 constrs. exhaust everything up to grading.

GELFAND-P000830

19/III - 79 Gelfand Sem.
Ogievetsky (Viktor Isaakovich)
Supergravity
(= gravity + supersymmetry)
$P_m,\ L_{mn}$ — generators of the Poincaré grp.
[At right:]

| | |
|---|---|
| $P_m$ | — 4 of them |
| $L_{mn}$ | — 6 of them |
| total | 10 of them |

$Q_\alpha, \overline{Q}_\alpha$ — spinor generators

$$\alpha = 1, 2$$

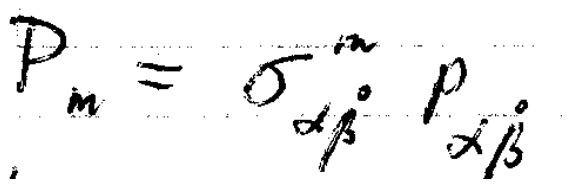


Original source details: GELFAND-P000830

**Uncertain reading.** The initial letters of “spinor” are retraced. The earlier strokes are illegible.

$$\{Q_\alpha, \overline{Q}_{\dot\alpha}\} = 2\sigma^m_{\alpha\dot\alpha} P_m; \tag{1}$$

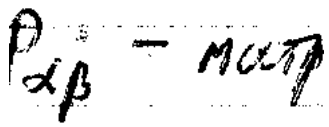
$$P_m = \sigma^m_{\alpha\dot\beta} P_{\alpha\dot\beta}$$



**Uncertain reading.** The upper index on $\sigma$ provisionally reads $m$, possibly $n$. The short stroke over $\alpha$ on the second $P$ resembles a detached tip of the letter. A dot is also possible.

Original source details: GELFAND-P000830

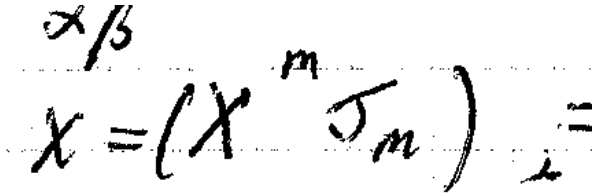
[At right:] $P_{\alpha\dot\beta}$ — matr. of ord. 2, corresp. to the vector $P$.



**Uncertain reading.** Faint marks above $\beta$ in the right-hand annotation may be the trace of a dot. A separate dot above $\alpha$ cannot be distinguished.

Original source details: GELFAND-P000830

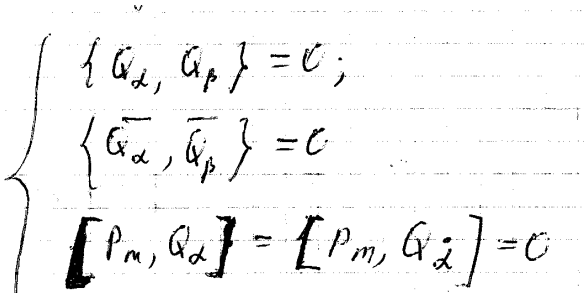
$$\underbrace{\mathbf{1}, \sigma_i}_{\substack{\text{matrs.}\\ \text{Pauli}}}; \qquad x = (X^m \sigma_m), = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}$$



**Uncertain reading.** The first letter provisionally reads $x$, possibly $\chi$. A short mark after the closing parenthesis is provisionally set as a comma.

Original source details: GELFAND-P000830

$$\begin{cases} \{Q_\alpha, Q_\beta\} = 0; \\ \{\overline{Q}_\alpha, \overline{Q}_\beta\} = 0 \\ [P_m, Q_\alpha] = [P_m, Q_{\dot\alpha}] = 0 \end{cases} \tag{1}$$



Original source details: GELFAND-P000830

**Uncertain reading.** The number (1) is repeated. There are no visible dots on the indices in the middle line and no visible bar over $Q_{\dot\alpha}$ in the last line.

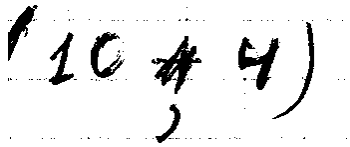
Thus a Lie superalgebra has been constructed whose even part is the Poincaré algebra.
Take 2 reps. of the Lorentz grp. $\left(\frac{1}{2}, 0\right),\ \left(0, \frac{1}{2}\right)$ (2-dimensional)
(it is $SL(2, \mathbb{C})$ 2 reps. — stand. and conjugate)
(Lorentz grp.: [crossed out: $A$] $A \to X^{+}AX, \quad X \in SL(2, \mathbb{C})$
$A$ — Hermitian $2 \times 2$, det — Lorentz metric.
we get a map into the Lorentz grp.)



Original source details: GELFAND-P000830

**Uncertain reading.** The crossed-out sign before the map resembles $A$.

Thus, 14-dimensional $(10, 4)$ — a superalgebra



Original source details: GELFAND-P000830

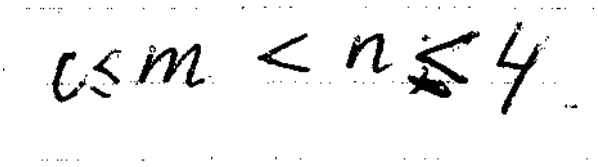
**Uncertain reading.** The sign between 10 and 4 is crossed out and resembles a plus. The comma is inserted below.

$L_{mn}$ skew-symm., there are [crossed out: ...] for $0 \leq m < n < 4$. [Below $L_{mn}$:] (since these are rotations)



Original source details: GELFAND-P000830

**Undeciphered passage.** The entry after "there are" is crossed out. Its original signs are illegible.



Original source details: GELFAND-P000830

**Uncertain reading.** The lower stroke in the last inequality is crossed out. The provisional reading is $< 4$. $\leq 4$ is also possible.

Suppose there are several spinor operators
Precisely: there are $\mathfrak{g}_0$ and $\mathfrak{g}_1$
$\mathfrak{g}_0$ — a Lie alg.; $\mathfrak{g}_1$ — $\mathfrak{g}_0$-module.
We want to constr. $\mathfrak{g}_0 \oplus \mathfrak{g}_1$. Write down the relations: in $\mathfrak{g}_1$ one needs to introduce.

GELFAND-P000831

[source number: 13]

In our case it is unique.

Now take several:

$$\left(\frac{1}{2},\frac{1}{2}\right) \qquad (1,0)+(0,1) \qquad \text{[crossed out: } \cancel{\left(\tfrac{1}{2},0\right)}\text{]},$$

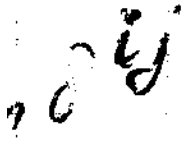


**Uncertain reading.** The crossed-out pair at the right provisionally reads $\left(\frac{1}{2},0\right)$.

Original source details: GELFAND-P000831

One can modify system (1).

$$\{Q^i_\alpha, \bar{Q}^j_{\dot\alpha}\} = 2\sigma^m_{\alpha\dot\alpha} P_m \delta^{ij} \tag{2}$$



**Uncertain reading.** The last factor is provisionally set as $\delta^{ij}$. The upper stroke is detached from the lower loop.

Original source details: GELFAND-P000831

$i, j$ — indices of internal symmetry.

We get a nontrivial mixing of internal symmetry with the space group —

proton + antiproton = momentum

(impossible in ordinary Lie algebras — there is a theorem)

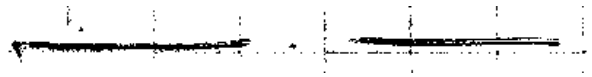


**Diagram and typed description.**

Original source details: GELFAND-P000831

Consider [crossed out: ...] supergravity — alg. (1).



**Undeciphered passage.** A short entry before “supergravity” is crossed out with dense diagonal strokes. Its letters are illegible.

Original source details: GELFAND-P000831

Realization (in Minkowski space):

$$P_m = -i\frac{\partial}{\partial x^m} \tag{3}_1$$

$$L_{mn} = -i(x_m\partial_n - x_n\partial_m), \qquad \partial_n = \frac{\partial}{\partial x^n}. \tag{3}_2$$

$Q$ is not realized in ordinary space

We want to extend the space. New variables:

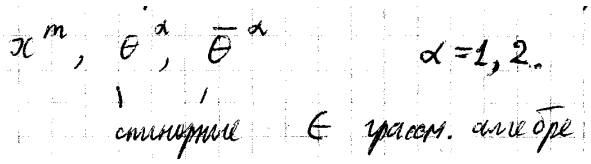


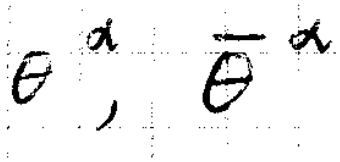
**Diagram and typed description.**

$x^m$, $\theta^\alpha$, $\bar\theta^\alpha$ $\alpha = 1, 2.$

spinorial $\in$ the Grassmann algebra

Original source details: GELFAND-P000831

$$\{\theta, \theta\} = 0$$



**Uncertain reading.** There is no discernible dot over the $\alpha$ on $\bar\theta$ in the list of variables.

Original source details: GELFAND-P000831

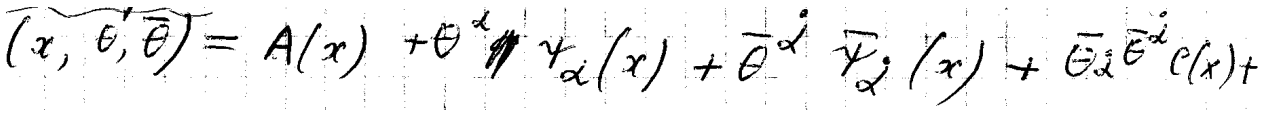
Functions on this space:

$$\begin{aligned}\varphi(x,\theta,\bar\theta) = A(x) + \theta^\alpha \dots \psi_\alpha(x) + \bar\theta^{\dot\alpha}\bar\psi_{\dot\alpha}(x) + \bar\theta_{\dot\alpha}\bar\theta^{\dot\alpha}C(x) + \\ + \theta^\alpha\theta_\alpha B(x) + \theta\sigma^\alpha\bar\theta h_\alpha(x) + \dots + \theta^\alpha\theta_\alpha\bar\theta_{\dot\beta}\bar\theta^{\dot\beta}D(x)\end{aligned} \tag{4}$$



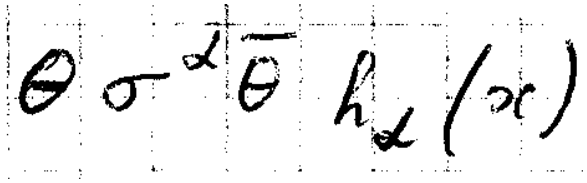
**Undeciphered passage.** A sign is crossed out between $\theta^\alpha$ and $\psi_\alpha(x)$. It may be $\varphi$ or $\psi$.

Original source details: GELFAND-P000831



Original source details: GELFAND-P000831

**Uncertain reading.** A + occurs both at the end of the first manuscript line and at the start of the second.



**Uncertain reading.** Both indices in $\theta\sigma^\alpha\bar\theta h_\alpha(x)$ are provisionally read as $\alpha$. The Latin $a$ is possible.

Original source details: GELFAND-P000831



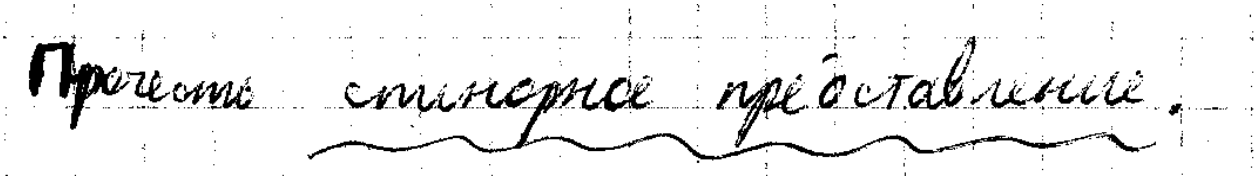
Original source details: GELFAND-P000831

**Uncertain reading.** The ellipsis before the last term is inserted from below with a caret.

There are 4 variables $\theta$.

Metric tensor $\varepsilon_{\alpha\beta}$: $\varepsilon_{11} = \varepsilon_{22} = 0, \qquad \varepsilon_{12} = -\varepsilon_{21} = 1$

Simplest spinor representation.



Original source details: GELFAND-P000831

**Uncertain reading.** The Russian word for “simplest” may be abbreviated. The wavy underline begins at “spinor”.

$$\begin{cases} Q_\alpha = -i\dfrac{\partial}{\partial\theta^\alpha} - (\sigma^m\bar\theta)_\alpha\overline{\cdots}\partial_m \\ \bar Q_{\dot\alpha} = i\bar\partial_{\dot\alpha} + (\theta\sigma^m)_{\dot\alpha}\partial_m \end{cases} \tag{5}$$



Original source details: GELFAND-P000831

**Undeciphered passage.** Writing is visible between $(\sigma^m\bar\theta)_\alpha$ and $\partial_m$ under a vertical zigzag. The original sign is illegible.

$\bar\partial_{\dot\alpha}$ — differentiation with respect to a Grassmann variable

GELFAND-P000832

This realizes the system of commutation relations (1).

Action of the group in superspace itself.

(the action on functions is given)

$$x'^m = x^m + c^m$$

under supertranslations:

$$\begin{cases} x^{m'} = x^m + i(\theta\sigma^m\bar\lambda - \lambda\sigma^m\bar\theta) \\ \theta^{\alpha'} = \theta^\alpha + \lambda^\alpha \\ \bar\theta^{\dot\alpha'} = \bar\theta^{\dot\alpha} + \bar\lambda^{\dot\alpha} \end{cases} \tag{6}$$



Original source details: GELFAND-P000832

**Uncertain reading.** The first $\theta$ on the right in the line for $x^{m'}$ has been retraced. It reads $\theta$. Any earlier sign is indiscernible.

$\lambda, \bar{\lambda}$ — infinitesimal shifts of the Grassmann variables.

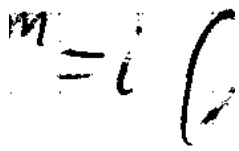
Perform the same transformation a second time with different parameters. We get a pure translation in $x^m$.

Set:

$$\delta_\lambda x^m = x^{m'} - x^m = i(\theta\sigma^m\bar{\lambda} - \lambda\sigma^m\bar{\theta})$$

Then

$$(\delta_\lambda\delta_\mu - \delta_\mu\delta_\lambda)x^m = i(\mu\sigma^m\bar{\lambda} - \lambda\sigma^m\bar{\mu}) \tag{7}$$



**Uncertain reading.** The right-hand side has $i$, with no visible $2$ preceding it.

Original source details: GELFAND-P000832

(i.e., a pure translation)

Transformation law of the superfield

$\psi'(x') = \psi(x)$ — the ordinary transformation law of a scalar field

For supertranslations of the (scalar) superfield:

$$\varphi'(x', \theta', \bar{\theta}') = \varphi(x, \theta, \bar{\theta}) \tag{8}$$

For the fields $A(x), \ldots$ from the expansion (4) this gives complicated formulas.

Consider some further new variables:

$$\begin{cases} x_L^m = x^m + i\theta\sigma^m\bar{\theta} \\ \delta x_L^m = 2i\theta\sigma^m\bar{\lambda} \end{cases} \qquad \delta\theta^\alpha = \lambda^\alpha \tag{9}$$

(there is only $\theta$ and no $\bar{\theta}$)

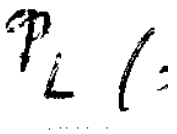
Special fields (**chiral**)



**Uncertain reading.** The middle letters in the Russian word for “chiral” are retraced. The earlier strokes there cannot be distinguished.

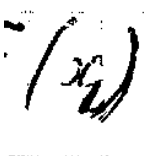
Original source details: GELFAND-P000832

$$\varphi_L(x_L, \theta) = A(x_L) + \theta^\alpha\psi_\alpha(x_L) + \theta^\alpha\theta_\alpha F(x_L) \tag{10}$$



**Uncertain reading.** The field symbol is provisionally set as $\varphi_L$. It may be $\Phi_L$.

Original source details: GELFAND-P000832



**Uncertain reading.** The final $x$ provisionally has subscript $L$. Its lower stroke touches the closing parenthesis.

Original source details: GELFAND-P000832

$$\begin{cases} x_R^m = x^m - i\theta\sigma^m\bar{\theta} \\ \delta x_{\underline{\underline{R}}}^m = -2i\lambda\sigma^m\bar{\theta} \end{cases} \qquad \delta\bar{\theta}^{\dot{\alpha}} = \bar{\lambda}^{\dot{\alpha}} \tag{11}$$



**Uncertain reading.** The $R$ in the line for $\delta x_R^m$ is underlined twice.

Original source details: GELFAND-P000832

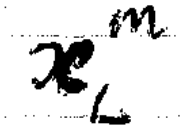
GELFAND-P000833

[source number: 14]

We obtain.

$$x^m = \frac{1}{2}(x_L^m + x_R^m)$$

$$x_L^m - x_R^m = 2i\theta\sigma^m\bar{\theta}$$



**Uncertain reading.** The initial $x$ in the second formula has been retraced. The earlier strokes are indiscernible.

Original source details: GELFAND-P000833



**Uncertain reading.** A faint trace to the right of $\bar{\theta}$ lies along the paper ruling. It may be a full stop.

Original source details: GELFAND-P000833



Original source details: GELFAND-P000833

**Diagram and typed description.**

. . . . . .

Divider of 26 dots. The fine horizontal stroke above the dots coincides with the paper ruling. Its relation to the divider is unclear.

GELFAND-P000834

26.3.79 Gelfand Sem.



**Diagram and typed description.**

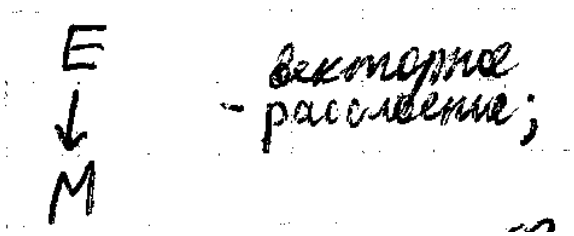
Overlapping horizontal and angular strokes appear below the date. No individual letters can be made out.

Original source details: GELFAND-P000834

Lebedev (Mitya)

On results of Ritt and Kirillov in differential algebra.

$M$ — a smooth manifold, dim $M = n$




Original source details: GELFAND-P000834

**Diagram and typed description.**

$E$ — vector bundle; $M$

$\Gamma^\infty(E)$ — smooth sections.

$$s_1, s_2 \in \Gamma^\infty(E)$$

Introduce a bilinear operation: $[\cdot, \cdot]$:

$$[s_1, s_2] \in \Gamma^\infty(E)$$

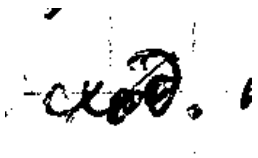


Original source details: GELFAND-P000834

**Uncertain reading.** The sign before $\Gamma^\infty(E)$ is retraced and provisionally read as $\in$.

a) bilin., skew-symm., Jacobi id.

b) cont. in the topol. of unif. conv. of derivs. on compact sets

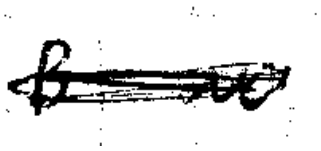


Original source details: GELFAND-P000834

**Uncertain reading.** The Russian abbreviation for convergence provisionally reads «схд.». A separate «о» may be present between «х» and the retraced «д».

c) $\operatorname{supp}[s_1, s_2] \subset \operatorname{supp} s_1 \cap \operatorname{supp} s_2$ (locality)

[crossed out: ...] This is called a differential Lie algebra.



Original source details: GELFAND-P000834

**Undeciphered passage.** The beginning of the crossed-out fragment has a loop resembling «В». The remaining letters are illegible.

GELFAND-P000835

[source number: 20]

The same definition in loc. coordinates

Let $F$ — the fiber; dim $F = 1$

$x_1, \dots, x_n$ — coords. on the base.

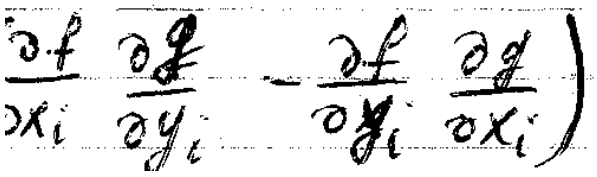
Then locally

$$[f, g] = \sum c_{k\ell}(x) \partial^k f \cdot \partial^\ell g, \qquad k, \ell \text{ — multi-indices} \tag{1}$$

Kirillov's main result: classification of such algebras.

Theorem 1. All loc. algebras with a fiber of dim. 1 are exhausted by two (essentially)

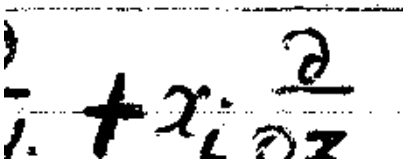
$$\text{a)} \quad M = \mathbb{R}^{2n}, \qquad [f, g] = \sum \left( \frac{\partial f}{\partial x_i} \frac{\partial g}{\partial y_i} - \frac{\partial f}{\partial y_i} \frac{\partial g}{\partial x_i} \right) \tag{2}$$



**Uncertain reading.** Overlapping strokes appear in the first numerator with $g$ and the second denominator with $y_i$. The earlier letter forms cannot be distinguished.

Original source details: GELFAND-P000835

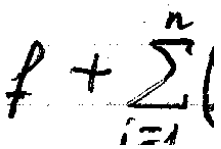
$$\text{b)} \quad M = \mathbb{R}^{2n+1}, \quad X_i = \frac{\partial}{\partial x_i}, \quad Y_i = \frac{\partial}{\partial y_i} + x_i \frac{\partial}{\partial z}, \quad Z = \frac{\partial}{\partial z} \text{ — vect. fields} \tag{3}$$



**Uncertain reading.** A plus is visible before $x_i \partial/\partial z$ in the entry for $Y_i$.

Original source details: GELFAND-P000835

$$[f, g] = fZg - gZf + \sum_{i=1}^{n} (X_i f \cdot Y_i g - X_i g \cdot Y_i f) \tag{3}_1$$



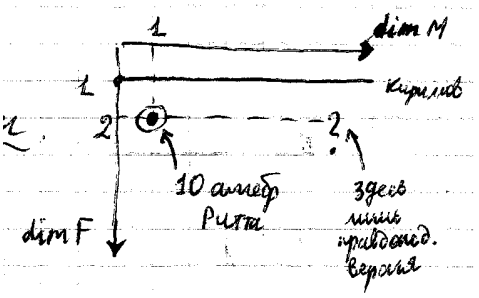
**Uncertain reading.** A plus is visible before the sum in $(3)_1$.

Original source details: GELFAND-P000835

All the others are formally isomorphic to them.

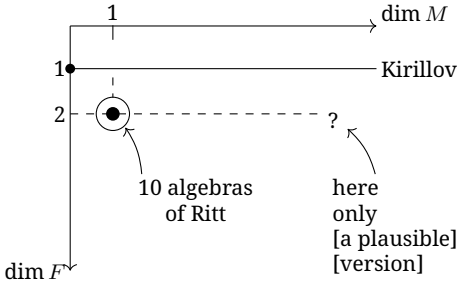
Ritt.

Differential Lie group and diff. Lie algebra.




Original source details: GELFAND-P000835

**Diagram and typed description.**




At the right margin. The last two lines at the right are provisionally read as «правдопод. / версия» (“plausible / version”).

Consider series

$$\sigma_1 = \sum_{i\geq 1} a_i \, a \, \xi^{-(i+1)} \tag{4}$$



Original source details: GELFAND-P000835

**Uncertain reading.** A separate dark sign resembling $a$ occurs between $a_i$ and $\xi$.

[crossed out: ...] and the commutator



Original source details: GELFAND-P000835

**Undeciphered passage.** A short entry at the left is crossed out. The underlying marks are illegible.

$$[\sigma_1, \sigma_2] = \sigma_1 \cdot \sigma_2 - \sigma_2 \cdot \sigma_1 \tag{5}$$

(here $\dim M = 1$, $\dim F = \infty$).

$[\xi, x] = 1$ — this determines everything.

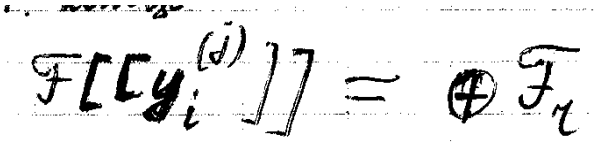


Original source details: GELFAND-P000835

**Diagram and typed description.**

$\mathcal{F}$ — diff. field of char. $0$, “$\partial = \frac{\partial}{\partial x}$”
$y_1(x), \dots, y_n(x)$ — diff. independent
Consider the ring

$$\mathcal{F}[[y_i^{(j)}]] \simeq \bigoplus \mathcal{F}_\tau, \qquad \tau \text{ — “order” — degree — number of factors} \tag{6}$$

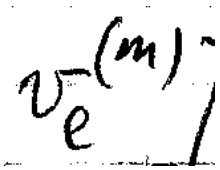


Original source details: GELFAND-P000835

**Uncertain reading.** The relation sign has a wavy upper stroke and is provisionally set as $\simeq$. The index is provisionally read as $\tau$.

Formal diff. Lie group ≡ form. group law ≡ set of fns.

$$U_i \in \mathcal{F}[[u_i^{(j)}, v_\ell^{(m)}]] \tag{7}$$



Original source details: GELFAND-P000835

**Uncertain reading.** A short horizontal stroke at the upper right of $v$ is joined to the letter. It may be a separate bar.

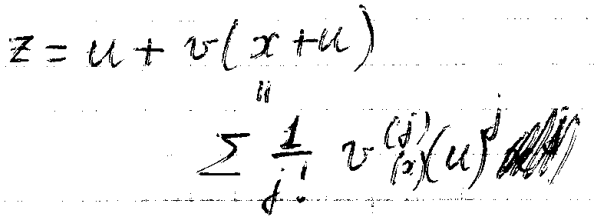
Satisfies 2 axioms:

a) $U_i(\bar{u}, 0) = \bar{u}_i, \qquad U_i(0, \bar{v}) = \bar{v}_i$

b) associativity.

GELFAND-P000836

Example.



Original source details: GELFAND-P000836

**Diagram and typed description.**
$z = u + v(x+u)$
$\| \; \| \; \| \; \

Form. series:

$$U = \sum_{k=0}^{\infty} P_k,$$

$$P_k = \text{homog. polyn. of deg. } k \text{ in } u_1, \ldots, u_n, u_1^{(1)}, \ldots, u_n^{(1)},$$

over a diff. field (a field of functions with diff.)



**Uncertain reading.** The start of the Russian word for “functions” is overwritten. Its earlier form is illegible.

Original source details: GELFAND-P000836

Theorem 2

Classific. of diff. groups with a fiber of dim. 2

$A^h$.

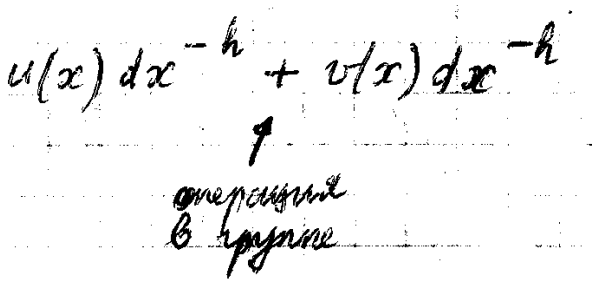


**Diagram and typed description.**

$u(x)\,dx^{-h} + v(x)\,dx^{-h}$

↑

operation
in the group

Original source details: GELFAND-P000836

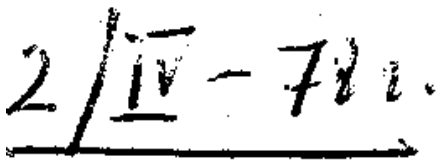
$$u \circ v = u + v$$

$$\Pi: \qquad u \circ v = u + v(x+u)$$

All groups are ext. of $\Pi$ by $A^h$.

GELFAND-P000837

2/IV–78 Gelfand Sem.



**Uncertain reading.** The last digit of the year is provisionally read as 8. It may be 9.

Original source details: GELFAND-P000837

K. Shabat. From the work of Griffiths and Chern.



**Uncertain reading.** The initial is provisionally read as the Russian «К.». It may be «Ю.».

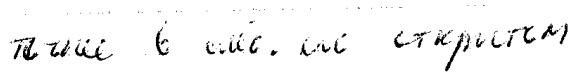
Original source details: GELFAND-P000837

A web — a finite number of foliations with transv. leaves. It is constructed on a manifold.
(net) — Web

$$X \subset P^n, \qquad \dim X = m, \qquad \deg X = d.$$

[Below $X$:] proj. set (algebr.)

A web in $\mathrm{Gr}(m, n-m)$, more precisely in some open subset of it:



Original source details: GELFAND-P000837

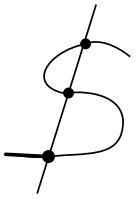
**Uncertain reading.** The Russian phrase between “in” and “open” provisionally reads «нек. его» (“some … of it”). The middle is faint.

$$U = \{H \mid H \cdot X = d \text{ distinct points}\} \quad \text{— an open set} \tag{1}$$

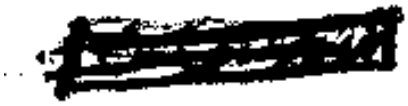


Original source details: GELFAND-P000837

**Diagram and typed description.**



([crossed out: ~~…~~] when there is multiplicity, discard it)



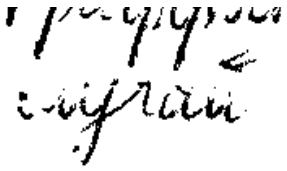
Original source details: GELFAND-P000837

**Undeciphered passage.** The first word inside the parentheses is crossed out. Its letters are indiscernible.

I.M.G.

Crofton’s formula: length of a curve = average number of points at which a line intersects it (average over lines with respect to an invariant measure).

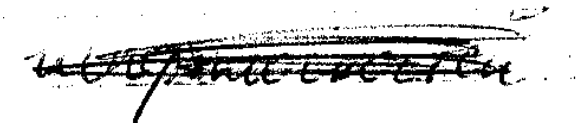
Griffiths and Chern, inspired by Penrose’s work, generalized this to the multidimensional [inserted below at left: ← case].



Original source details: GELFAND-P000837

**Uncertain reading.** The word “case” is below the line at the left. A short arrow above it points left.

The leaf passing through the point $H_0 \in U$, consists of [crossed out: ~~surfaces~~].

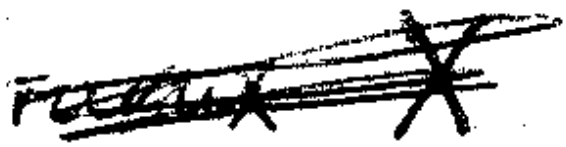


Original source details: GELFAND-P000837

**Uncertain reading.** The crossed-out Russian word is provisionally read as «поверхностей» (“surfaces”).

[crossed out: ~~points of $X$~~] planes

$$\{H \in U \mid \exists i \in \{1, 2, \ldots, d\};\ P_i \in H\}.$$

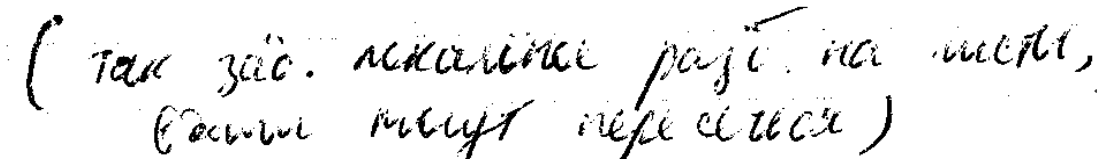


Original source details: GELFAND-P000837

**Uncertain reading.** “Points of $X$” is crossed out at the start of the line. There is a short stroke below the first $P_i$.

$$H_0 \cdot X = P_1 + \ldots + P_d$$
$$\{H \in U \mid \exists i \in \{1, 2, \ldots, d\},\ P_i \in H\} \tag{2}$$

This is a foliation of codimension 1. (so here the planes are div. into leaves, they themselves can intersect)



Original source details: GELFAND-P000837

**Uncertain reading.** The Russian abbreviations provisionally read «зд.» (“here”) and «разд.» (“div.”).

I.M.G. [crossed out: ~~Questions~~]



Original source details: GELFAND-P000837

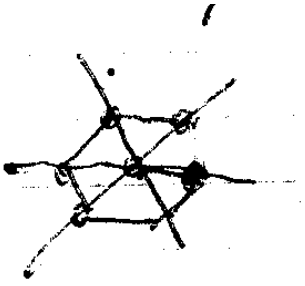
**Uncertain reading.** The crossed-out word provisionally reads «Вопросы» (“Questions”).

Problem: there are 3 foliations (transv.) When can they be straightened? [Inserted below “there are 3 foliations”:] in the plane.

(2 always!!)

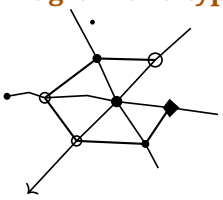
3 can be straightened $\Leftrightarrow$ it is hexagonal, i.e.

$\approx 1927$



Original source details: GELFAND-P000837

**Diagram and typed description.**



The redraw follows the visible lines, small circles and thickened nodes. The order of traversal and the direction of some small arrows are indistinct.

— such a path (infinitesimal) must be closed. — something like curvature.



**Diagram and typed description.**



Original source details: GELFAND-P000837

GELFAND-P000838

[source number: 29]

Now $m = 1$.

$$X \subset P^n$$
$$|$$
proj. curve

$\check{P}^n$ — dual proj. space.

$$\check{\mathbb{P}}^n \supset U = \{H \mid H \circ X = d \text{ distinct points}\} \tag{3}$$



**Uncertain reading.** A small oval is visible between $H$ and $X$, provisionally set as $\circ$.

Original source details: GELFAND-P000838

On $U$ we get a web of codimension 1.

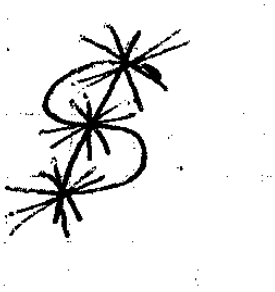
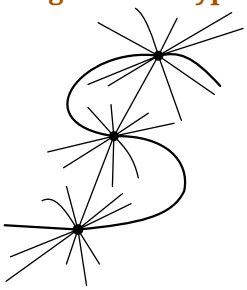


**Diagram and typed description.**



Original source details: GELFAND-P000838

The redraw follows the three centers and the discernible strokes. Some strokes merge, so their exact number is uncertain.

Abelian eq. for the web.

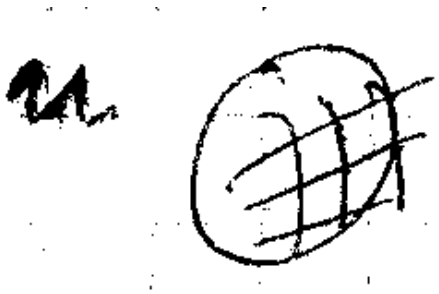
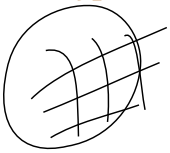


**Diagram and typed description.**

$[u]$



Original source details: GELFAND-P000838



**Uncertain reading.** The sign to the left of the oval is provisionally set as $u$. It may be $U$.

Original source details: GELFAND-P000838

$$\begin{cases} u_1 = \text{const.} \\ \vdots \\ u_n = \text{const} \end{cases} \tag{4}$$

The Ab. eq. for the web has the form:

$$\sum_{i=1}^{d} f_i(u_i)\, du_i = 0, \qquad \sum_{i=1}^{d} f_i(u_i)\, du_i \in \Omega^1(U) \tag{5}$$

There is a dependence among the normal vectors (cotang. vectors), and the coefficients have the form $f_i(u_i)$.

Rank of the web $\underset{\text{def}}{=}$ max. number of lin. independent Abelian eqs., that it satisfies.

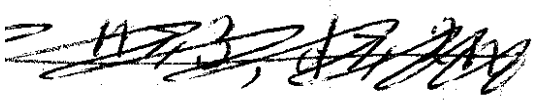
Prop. 1 The rank of the web constructed from a curve of genus $g$, equals $g$.

[crossed out: ~~Another theorem 1 — Under the conditions~~ ~~$n \geq 3,\ d \geq 2n$~~]



**Uncertain reading.** The first crossed-out word provisionally reads «Еще» (“Another”).

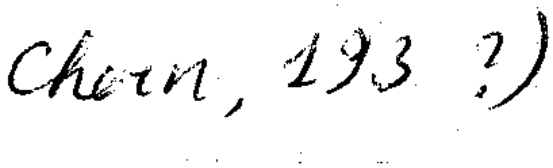
Original source details: GELFAND-P000838



**Uncertain reading.** The crossed-out conditions provisionally read $n \geq 3,\ d \geq 2n$.

Original source details: GELFAND-P000838

Theorem 1 (Chern, 193 ?)



**Uncertain reading.** A question mark follows 193 in the manuscript.

Original source details: GELFAND-P000838

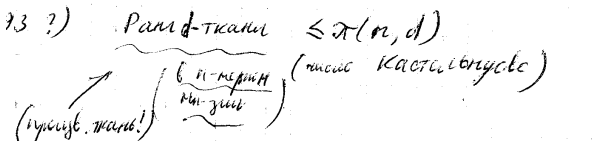


**Diagram and typed description.**

Rank of the $d$-web $\leq \pi(n,d)$ (Castelnuovo number)

(in an $n$-dim. set)

(arbitr. web!)

Original source details: GELFAND-P000838

$$\pi(n,d) \underset{\text{def}}{=} \text{max. genus of a curve of degree } d \text{ in } \mathbb{P}^n, \qquad n, d \geq 2. \tag{6}$$

There is a short formula

$$\pi(n,d) = m\left(d - \frac{1}{2}(m+1)(n-1) - 1\right), \quad \text{where } m = \left[\frac{d-1}{n-1}\right]. \tag{7}$$

GELFAND-P000839

Theorem 2.

(main)

If $n \geq 3,\ d \geq 2n$ and the $d$-web has rank $\pi(n,d)$,

then

1) It can be linearized

2) It extends to a web obtained from $X \subset \mathbb{P}^n$.

The connection between such eqs. and Abelian differentials is explained.

Reference:

(Sonderdruck aus) Jber. d. Dt. Math.-Verein, 80 (1978), 13–110

[Under «Sonderdruck aus»: offprint from.]

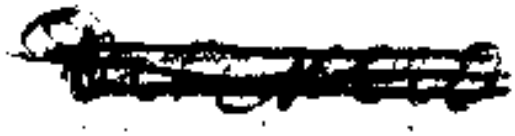
(Jahresberichte der Deutsche[...] Math. Vereinigung)



Original source details: GELFAND-P000839

**Uncertain reading.** The ending after “Deutsche” is uncertain. “Deutschen” and “Deutscher” are possible.

[crossed out: ...]



Original source details: GELFAND-P000839

**Undeciphered passage.** A word below the reference is crossed out. Its initial loop is visible, and the remaining letters are obscured.

A geometry is being developed for

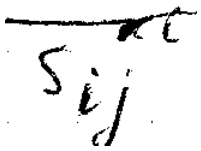


Original source details: GELFAND-P000839

**Diagram and typed description.**
projective connections.

The left stroke has a left-pointing arrow. A separate stroke runs under the words at right.

Zamolodchikov A. (Sasha) Suppose there is $S_{ij}^{k\ell}(\lambda)$



Original source details: GELFAND-P000839

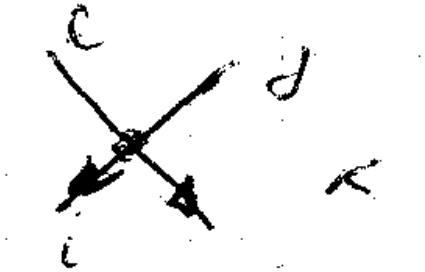
**Uncertain reading.** The upper indices are provisionally read as $k, \ell$. They touch the stroke from the preceding line.

Suppose there is also a picture:
$L$ lines, angles $\lambda_1, \ldots, \lambda_L$
$\lambda_i$ in increasing order
(from some line
[crossed out: ~~Then~~]



Original source details: GELFAND-P000839

**Uncertain reading.** The crossed-out word provisionally reads «То» (“Then”).



Original source details: GELFAND-P000839

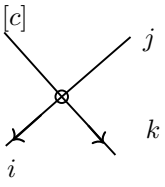
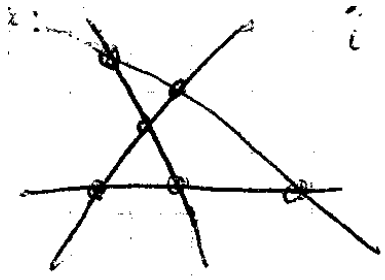


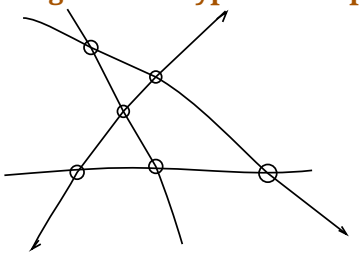

**Diagram and typed description.** The upper-left sign resembles $c$ and may be $\ell$.

indices $= 1, 2, \ldots, N$



Original source details: GELFAND-P000839

**Diagram and typed description.**

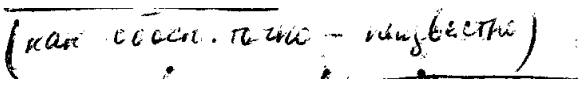


There are six circled intersections. Short strokes at the ends are drawn as visible. Some arrow directions are unclear.

To a node — a matrix; to tails — indices, I sum over the links.
We get a matrix (large). What is its behavior as $L \to \infty$?

$$Z^{j_1 \ldots j_L}_{i_1 \ldots i_L}(\lambda_1, \ldots, \lambda_L).$$

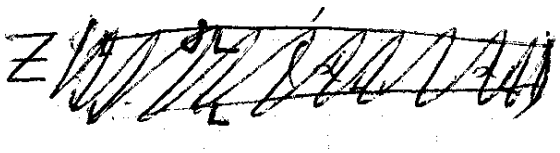
Problem 1. There is an asymptotic formula
(how to calc. it exactly — unknown)



Original source details: GELFAND-P000839

**Uncertain reading.** The Russian words provisionally read «сосч. точно» (“calculate exactly”). «сосч. её» (“calculate it”) is possible.

$$\ln Z_{\dots} \sim \text{proport. to the number of nodes} \tag{1}$$



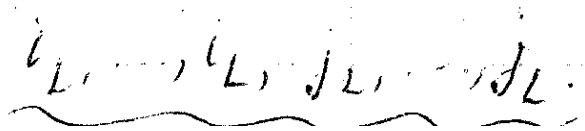
**Uncertain reading.** After $Z$ the hatching covers signs resembling the indices $j_1 \dots j_L$, $i_1 \dots i_L$ and arguments $(\lambda_1, \dots, \lambda_L)$. Some of the signs cannot be made out.

Original source details: GELFAND-P000839

$$Z \underset{\text{df}}{=} Z^{i_1 \dots i_L}_{i_1 \dots i_L} \qquad (\text{“trace”}) \qquad (\text{sum over repeated indices}) \tag{2}$$

Each line $\rightsquigarrow$ parameter $\lambda_j$

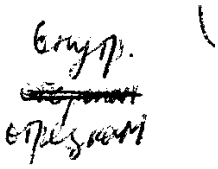
(the rays are numbered clockwise $i_1, \dots, i_L, j_1, \dots, j_L$.)



**Uncertain reading.** The first subscripts on $i$ and $j$ are provisionally set as the digit 1. Their shapes also resemble $L$.

Original source details: GELFAND-P000839

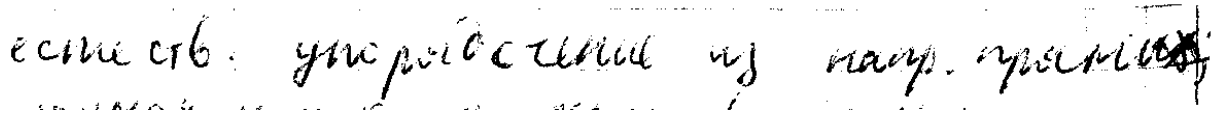
$$Z^{j_1 \dots j_L}_{i_1 \dots i_L} = \sum_{\substack{\text{internal} \\ \text{segments}}} \left( \prod_{\substack{\text{points of} \\ \text{intersection}}} S^{**}_{**}(\lambda_a - \lambda_b) \right) \tag{3}$$



**Undeciphered passage.** The middle line below the sum is crossed out. The word is illegible.

Original source details: GELFAND-P000839

[At right below the formula:] natural ordering from the dir. of the lines



Original source details: GELFAND-P000839

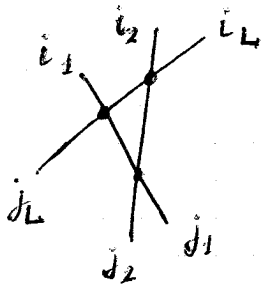
**Uncertain reading.** The ending of the Russian word for “lines” is overwritten. The original strokes are indiscernible.

there is a horiz. line that they cross at the very bottom.

There are fns. that do not change under parallel translation.

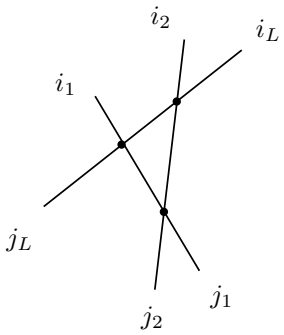
GELFAND-P000840

[source number: 30]

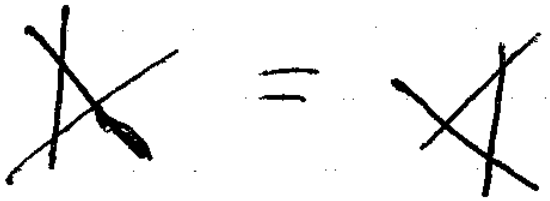


Original source details: GELFAND-P000840

**Diagram and typed description.**

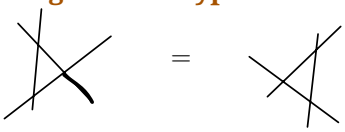



Associat.: $Z$ does not depend on the relative arrangement of the lines.
It suffices for it to hold for 3 lines:



Original source details: GELFAND-P000840

**Diagram and typed description.**



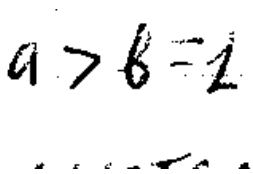
The thickened part of the descending line at left may be a retraced stroke or an arrow.

Prop. 2. such matrices exist.
On asymptotics:

$$\ln Z^{j_1 \dots j_L}_{i_1 \dots i_L} \sim L^2 \qquad \text{as } L \to \infty \tag{4}$$

(if without associativity) and the asympt. does not depend on the choice of indices $i$ and $j$.
[Under «asympt.»: (leading term)]
Prop. 3. Asymptotics as $L \to \infty$

$$\ln Z(\lambda_1, \dots, \lambda_L) = \sum_{a>b=1}^{L} f(\lambda_a - \lambda_b) + \text{remainder term} \tag{5}$$

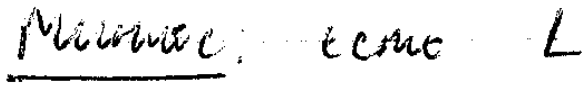


Original source details: GELFAND-P000840

**Uncertain reading.** The lower limit reads "$a > b = 1$." The upper limit is $L$.

(free en. of a large system = sum of pairwise

Opinion: there are $L$ kinds of particles; they move along straight lines and collide; on colliding they retain their velocities, but change color



**Uncertain reading.** The heading provisionally reads «Мнение» (“Opinion”).

Original source details: GELFAND-P000840



**Uncertain reading.** The opening strokes of the Russian word for “velocities” are overwritten. The earlier letterform is indistinguishable.

Original source details: GELFAND-P000840

[Under «color»: (kind)]

Here $Z$ — $S$-matrix;

There is an interpr. in which $Z$ — the partition function

The weights of $S$ — exp of the potential. (probabilities?)

There are initial and final colorings $i_1, \dots, i_L$

$$j_1, \dots, j_L.$$

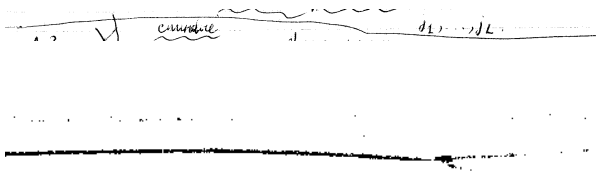
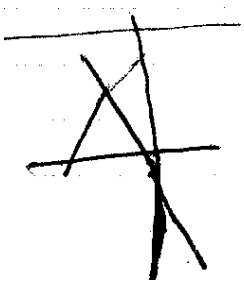


**Diagram and typed description.**

Original source details: GELFAND-P000840

A. Z.



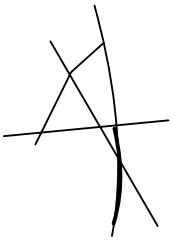
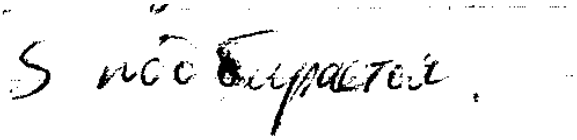


**Diagram and typed description.** Two crossings near the horizontal line merge in a thickened stroke. The long divider is visible on the excerpt at left.

Original source details: GELFAND-P000840

spin variables — in polygons; nearest-neighbor interaction;

$S$ is chosen.



**Uncertain reading.** The short note provisionally reads «подбирается» (“is chosen”). The opening letters and the heavy stroke remain unclear.

Original source details: GELFAND-P000840

$$Z = \sum_{\{\sigma\}} e^{-\beta H} \qquad H = \sum h(\sigma_A, \sigma_B)$$

Exactly calculable partition functions — all of this kind!

For the Ising model:

$$S_{ij}^{k\ell} = \frac{ijk\ell + 1}{2} e^{-\frac{\beta}{2}(i+j+k+\ell)}$$

(after passing to the dual lattice)

GELFAND-P000841

24.9.79 Gelfand Sem.

Renata Ernestovna KALLOSH on the conf. in Cambridge

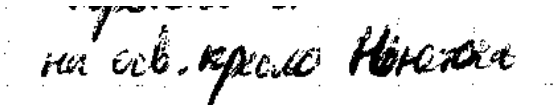
(physicist)

(month-long working meeting on quantum grav., approx. 40 people)

20 people from the USA; including Chern, Singer, Milnor, Penrose;

Hawking, De Witt, Sciama, Coleman, Jackiw

[Below Hawking, with a short vertical stroke:] candidate for Newton's vacant chair



**Uncertain reading.** The abbreviation preceding "chair" is provisionally read «св.». The translation takes it as "vacant."

Original source details: GELFAND-P000841

Gibbons — physicist, works with Hawking — talk on quantum grav.



**Uncertain reading.** The abbreviation before "on quantum grav." provisionally reads «докл.» ("talk"). «сем.» ("seminar") is possible.

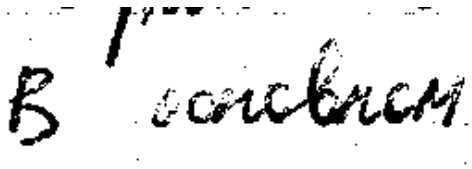
Original source details: GELFAND-P000841

Idea: one must take account of the effects of nontriv. topology of space/time on small scales.

(many little "handles").

The theory has yet to be constructed, but there has been signif. progress.

Mainly Hawking:

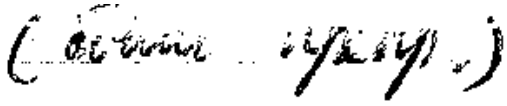


**Uncertain reading.** The Russian before Hawking provisionally reads «В основном» ("Mainly"). The opening strokes of the second word are faint.

Original source details: GELFAND-P000841

1) Space-time foam (foam) — published in Nuclear Physics

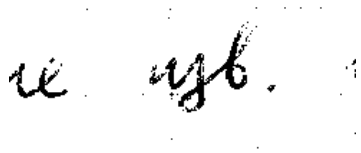
(ordinary journ.)



**Uncertain reading.** The note below the journal title is provisionally read «обычн. журн.» ("ordinary journ.").

Original source details: GELFAND-P000841

2) Page, Atiyah  something in the field of known gravitons



**Uncertain reading.** The abbreviation provisionally reads «изв.» (“known”). «изл.» (“emitted”) is possible.

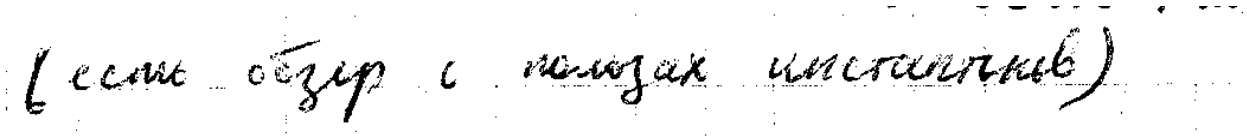
Original source details: GELFAND-P000841

(from the group of Hawking)

3) Coleman, Jackiw  — on grav. taking topol. numbers into account —

on Yang–Mills.

(there is a survey on instanton methods)



Original source details: GELFAND-P000841

**Uncertain reading.** The preposition before “methods” provisionally reads «о» (“on”). The letter is open and resembles «с».

4) Milnor — classif. of 4-dimensional manifolds

Someone on bundles on 4-manifolds.

5) Singer — on the signature of a 4-manifold.

On supermanifolds

Considered a route to quantization

De Witt is writing a book on supermanifolds

Singer is giving lectures



**Diagram and typed description.**

Original source details: GELFAND-P000841

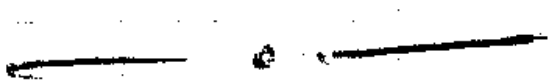
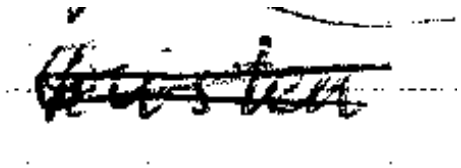


**Diagram and typed description.**

Original source details: GELFAND-P000841

Something obtained independently by the speaker and by someone from the conf.

[At right:] [Crossed out: ~~Christen[...]~~] Christensen, Duff;



**Uncertain reading.** The crossed-out name provisionally reads “Christen[...].”

Original source details: GELFAND-P000841

Kallosh — the speaker

Consider the eq.

$$\Delta[A,B]\,\Phi[A,B]=0 \tag{1}$$

Here

$$\phi[A,B]=\Psi_{\alpha_1\ldots\alpha_{2A},\dot\alpha_1,\ldots,\dot\alpha_{2B}}(x)$$

— a multispinor, transforms under the group $SO(4)$



Original source details: GELFAND-P000841

**Uncertain reading.** The second letter of the group name is open. It is provisionally set as $SO(4)$.

$x\in$ [Crossed out: ~~...~~] $R^4$.



Original source details: GELFAND-P000841

**Undeciphered passage.** One sign is crossed out before $R^4$. Its original form is indistinguishable.

$\Delta$ — some operator; at one time from Hodge–de Rham.

there is also the half Dirac here

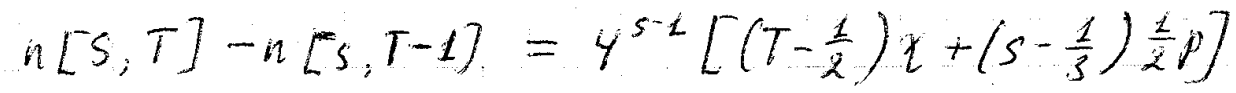


Original source details: GELFAND-P000841

**Uncertain reading.** The abbreviation before “Dirac” is provisionally read «пол.» and rendered “half.”

$$S=A+B,\qquad T=A-B.$$

$$n[S;T]-n[s;T-1]=4^{s-1}\left[\left(T-\frac{1}{2}\right)\chi+\left(s-\frac{1}{3}\right)\frac{1}{2}p\right] \tag{2}$$



Original source details: GELFAND-P000841

**Uncertain reading.** The $S/s$ forms in (2) vary in size. Their case is uncertain in places.

[Below $n$, with a short vertical stroke:] number of zero modes

(dim. of the solution space, with spin $A$ and $B$);

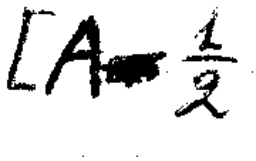
$\chi,p$ — Euler and Pontryagin numbers

GELFAND-P000842

[source number: 66]

Here there are the Gauss–Bonnet and Atiyah–Singer theorems, and so on.

$$n[A,B]-2n\left[A\overset{?}{-}\frac{1}{2},B\right]=4^{A+B-1}\left(A-\frac{7}{12}\right)\left(\chi+\frac{1}{2}p\right). \tag{3}$$

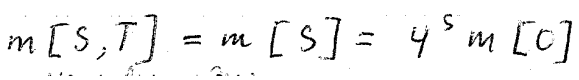


**Uncertain reading.** The sign before $1/2$ is heavily inked over. A minus is a possible reading.

Original source details: GELFAND-P000842

On nonzero modes:

$$m[S;T]=m[S]=4^{s}m[0] \tag{4}$$



**Uncertain reading.** The case of $s/S$ in the exponent is unclear. The last argument is provisionally set as $0$.

Original source details: GELFAND-P000842

[Under $m[S;T]$:] nonzero modes:

$$\Delta\Psi_i=\lambda_i\Psi_i \tag{5}$$

In supergravity the nonzero modes cancel and only the contribution of the zero ones remains.

(3) [Crossed out: ~~...~~] only if the manifold is self-dual.



**Undeciphered passage.** The word is obscured by heavy crossing-out.

Original source details: GELFAND-P000842

Hawking proved something earlier.

Coeff. in heat prop.



**Uncertain reading.** The Russian abbreviation is provisionally read «теплопров.», rendered “heat prop.”

Original source details: GELFAND-P000842

$$B_4 = n + m$$

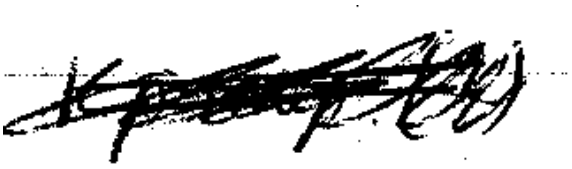
They proved that the funct. integral has no divergences.
(in gravity)
The speaker did something too.
Interesting problem: the same on a supermanifold.
Singer is working on this now.



**Diagram and typed description.**

Original source details: GELFAND-P000842

The group $SO(8)$, came up, something was wrong.
French group: [Crossed out: ... $(8)$] symmetry $E_7$ and



**Undeciphered passage.** The crossed-out letters before the visible "(8)" cannot be read confidently.

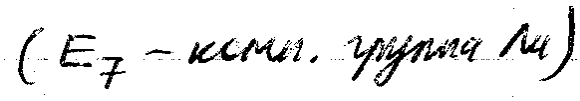
Original source details: GELFAND-P000842

$$E_7 \longrightarrow SU(8)$$

then ext. supergravity

$$\dim E_7 = 133$$

($E_7$ — comp. Lie group)



**Uncertain reading.** The abbreviation before "Lie group" provisionally reads «комп.» ("comp.").

Original source details: GELFAND-P000842

Interesting
[Crossed out: ...] to make a super index theorem



**Undeciphered passage.** A mark before "to make" is inked out. Its original form is indistinguishable.

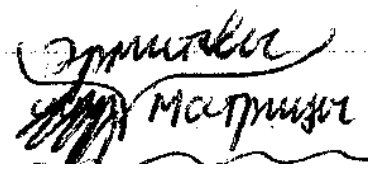
Original source details: GELFAND-P000842

Remarks:

$F_4$

[Inserted above the line:] Hermitian

[Crossed out: ...] matrices of order 3 made of octonions; in them one constructs a Jordan add..



Original source details: GELFAND-P000842

**Undeciphered passage.** A passage before “matrices” is crossed out. “Hermitian” is inserted above it.



Original source details: GELFAND-P000842

**Uncertain reading.** The last abbreviation is provisionally read «слож.» (“add.”).

product $\frac{1}{2}(AB + BA)$.

one must take the automorphism group of this Jord. algebra. We get $F_4$

Chern — adviser to the Chin. president — works in Chicago.

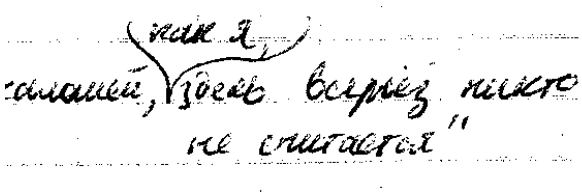
Yang gives lectures in China to an audience of a thousand.

I. M. G. — what physicists said about Einstein.

a) E. and N. Bohr are the greatest phys. of the century

b) E. did splendid work in his youth, but then spent 30–40 years on nonsense (tried to create a unified field theory).

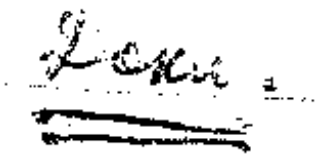
c) E. to Born: “I want to invite you to Princeton, but an old nag, [Inserted above the line: like me,] no one takes seriously here.”



Original source details: GELFAND-P000842

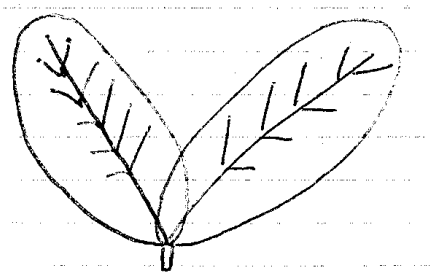
**Uncertain reading.** The end of the quotation reads «никто не считается». An angled mark leads to the insertion “like me.”

[In the left margin:] Blackboard.



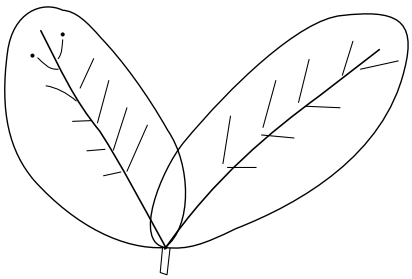
Original source details: GELFAND-P000842

**Uncertain reading.** The last letter of the Russian word is provisionally read «а».



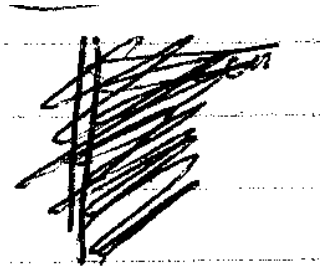
Original source details: GELFAND-P000842

**Diagram and typed description.**



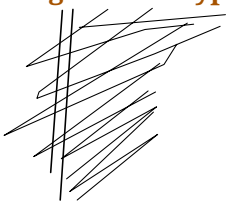
The two outlines meet at the base. At right, some short branches are detached from the main line.

connection of theor. physics with mathematics:
in fundamental matters
(Yang's drawing)



Original source details: GELFAND-P000842

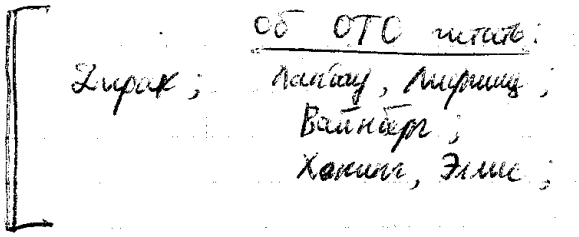
**Diagram and typed description.**



Near-vertical lines and diagonal hatching. The relationship to the drawing at left is unclear.

GELFAND-P000843

V. E. Zakharov.




Original source details: GELFAND-P000843

**Diagram and typed description.**

On general relativity, read:

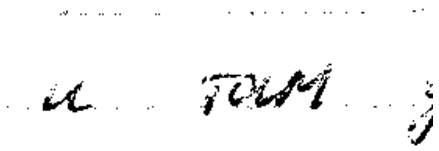
Dirac; Landau, Lifshitz;
Weinberg;
Hawking, Ellis;

In the right margin. A left bracket groups the list.

Mathematical problem:

$$k = (k_1, \ldots, k_N)$$

Consider the fn. $\omega(k)$ (real) and the space of pairs $(k_1, k_2)$ of dimension $2N$ and there the eq.




Original source details: GELFAND-P000843

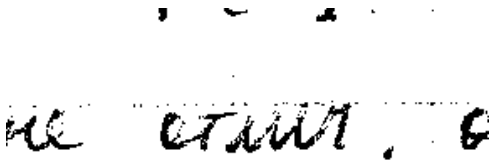
**Uncertain reading.** The short Russian word is provisionally read «там» ("there").

$$\omega(k_1 + k_2) = \omega(k_1) + \omega(k_2) \tag{1}$$

$\omega$ — nonlin. $\Rightarrow$ we get a surface $\Gamma$ in $R^{2n}$.
Is there another fn. $f$, satisfying the same eq. on $\Gamma$:

$$f(k_1 + k_2) = f(k_1) + f(k_2)$$

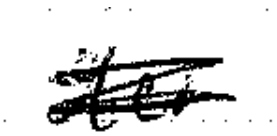
and which does not differ from $\omega$ by a linear one?



Original source details: GELFAND-P000843

**Uncertain reading.** The abbreviated verb after the negation is provisionally read «отлич.» (“differ”).

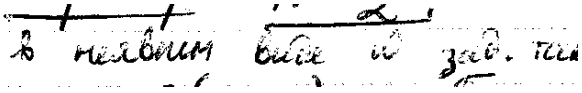
[Crossed out: ~~This~~]



Original source details: GELFAND-P000843

**Uncertain reading.** The crossed-out word provisionally reads «Это» (“This”).

This sit. is exceptional because the dimension is large.
Idea of Zakh. and Shulman:
how to get all eqs. having infinitely many first integrals.
Example: $N = 2$.
We will specify it by means of a parametrization, $\vec{k} = (p, q)$;
[In small writing below the line:] implicitly $\omega$ is given as follows:



Original source details: GELFAND-P000843

**Uncertain reading.** The small Russian words provisionally read «в неявном виде» (“implicitly”).

$$\omega(p, q), \qquad \begin{cases} p = \xi_1 - \xi_2 \\ q = a(\xi_1) - a(\xi_2) \\ \omega = b(\xi_1) - b(\xi_2) \end{cases} \tag{2}$$

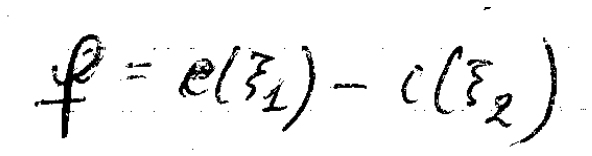
Another [Crossed out: ~~...~~] fn. $f$:



Original source details: GELFAND-P000843

**Undeciphered passage.** A short entry before “fn.” is crossed out. The complete word is illegible.

$$\begin{cases} p = \xi_1 - \xi_2 \\ q = a(\xi_1) - a(\xi_2) \\ f = c(\xi_1) - c(\xi_2) \end{cases} \qquad c \text{ — another fn. (arbitrary)} \tag{3}$$

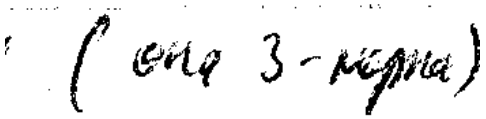


**Uncertain reading.** Provisionally set as $f = c(\xi_1) - c(\xi_2)$. The letter at left also resembles $\varphi$, and the first at right resembles $e$.

Original source details: GELFAND-P000843

Question: is this a general construction?

Idea: we will give a parametrization of $\Gamma$ (it is 3-dimensional)



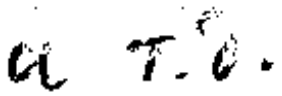
**Uncertain reading.** The Russian endings provisionally read «она 3-мерна» (“it is 3-dimensional”).

Original source details: GELFAND-P000843

$$\begin{cases} p_1 = \xi_1 - \xi_2 \\ q_1 = a(\xi_1) - a(\xi_2) \\ p_2 = \xi_2 - \xi_3 \\ q_2 = a(\xi_2) - a(\xi_3) \end{cases} \tag{4}$$

$$\underbrace{p_1 + p_2}_{p_3} \qquad \underbrace{q_1 + q_2}_{q_3} \qquad p_3 = \xi_1 - \xi_3$$

and so on.



**Uncertain reading.** The opening sign is provisionally read «и» (“and”).

Original source details: GELFAND-P000843

GELFAND-P000844

[source number: 67]

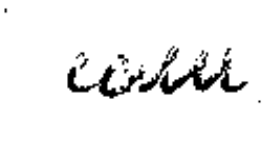
[Above the first line crossed out: ~~...~~]



**Undeciphered passage.** A short entry above the first line is inked out. The letters are indistinguishable.

Original source details: GELFAND-P000844

Hence if degenerate, then serious (functional parameter).



**Uncertain reading.** The short Russian word is provisionally read «если» (“if”).

Original source details: GELFAND-P000844

There is a method of constr. systems with a Lax repr. (Zakharov–Shabat)

$$\frac{\partial L}{\partial t} - \frac{\partial A}{\partial y} + [L, A] = 0 \tag{5}$$

$$L = \frac{\partial^n}{\partial x^n} + \dots$$
$$A = \frac{\partial^m}{\partial x^m} + \dots$$

(5) — nonlin. syst. of eqs. for the coeffs.

Consider its linearization.

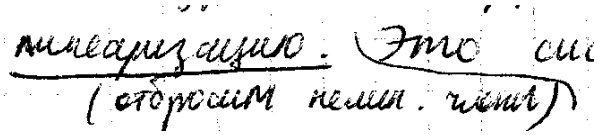


**Diagram and typed description.**
linearization. This

(discard the nonlin. terms)

A note below the line. The connecting stroke reaches “This.”

Original source details: GELFAND-P000844

This is a syst. with const. coeffs., it is solved by the method of Fourier.

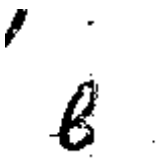
$$u_j \sim \exp(-\omega t + qy + px)$$
$$\omega = \omega(p, q)$$

Those syst. look as in



**Uncertain reading.** The opening Russian word provisionally reads «Те» (“Those”).

Original source details: GELFAND-P000844



**Uncertain reading.** After “as” there is a separate «в» (“in”). The sentence breaks off before the symbols.

Original source details: GELFAND-P000844

$a(\xi)$ — symbol of the oper. $L$

$b(\xi)$ — ″ — $A$

[In the right margin:]

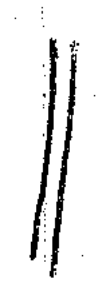


**Diagram and typed description.**

$$L = L_0 + \delta L$$
$$A = A_0 + \delta A$$

Original source details: GELFAND-P000844

$\delta L, \delta A$ — everything that contains variable coeffs.

Degenerate disp. laws are connected with the inverse problem.

$$p = \xi_1 - \xi_2$$
$$q = \ell(\xi_1) - \ell(\xi_2)$$
$$\omega = a(\xi_1) - a(\xi_2)$$

$a, \ell$ — symbols of $A_0, L_0$.

Consider the G.–L.–M. eq.:

$$F + K + K \oplus F = 0$$



**Uncertain reading.** A circled plus is visible between $K$ and $F$.

Original source details: GELFAND-P000844

$$\frac{\partial F}{\partial t} + [L_0, F] = 0$$
$$\frac{\partial F}{\partial y} + [A_0, F] = 0$$

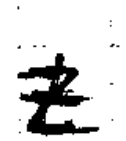


**Uncertain reading.** The plus on the second line is retraced and still visible.

Original source details: GELFAND-P000844

$$K = -F$$

$$F = \exp\big(\xi_1 x - \xi_2 z + [\ell(\xi_1) - \ell(\xi_2)]y + [a(\xi_1) - a(\xi_2)]t\big)$$



**Uncertain reading.** The letter after $\xi_2$ is heavily overwritten. It provisionally reads $z$.

Original source details: GELFAND-P000844

Here the connection is estab. when $\ell$ and $a$ — polynomials. One can estab. a connection for rat. functions.



**Uncertain reading.** The letter $\ell$ before “and” has been retraced.

Original source details: GELFAND-P000844

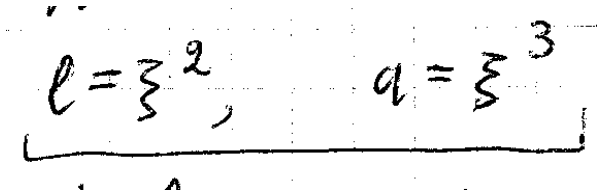


**Diagram and typed description.**
$\ell = \xi^2, \qquad a = \xi^3$

Original source details: GELFAND-P000844

$L, A$ are to be found — we get KdV.



**Uncertain reading.** The Russian verb provisionally reads «ищутся» (“are to be found”).

Original source details: GELFAND-P000844

GELFAND-P000845

Integr. by the inverse problem $\Rightarrow$ the disp. law is degenerate.

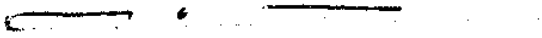


**Diagram and typed description.**

Original source details: GELFAND-P000845

Matrices $L$ and $A$: no longer necessarily

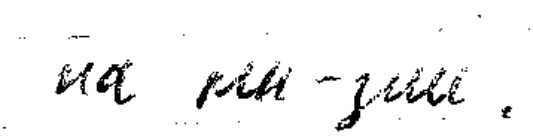
[In the left margin:] Altern.



**Uncertain reading.** The underlined marginal word is provisionally read «Альтерн.» (“Altern.”).

Original source details: GELFAND-P000845

If the eq. is integr. and the disp. law is nondeg., then the interaction $\equiv 0$ on the manifold.



**Uncertain reading.** The final Russian abbreviation provisionally reads «мн-зии» (“manifold”).

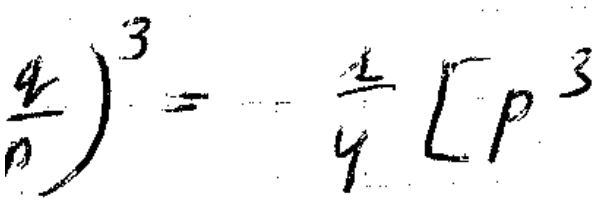
Original source details: GELFAND-P000845



**Diagram and typed description.**

Original source details: GELFAND-P000845

$$\omega = \frac{1}{8}\left[(p+\tfrac{q}{p})^3 - (-p+\tfrac{q}{p})^3 = \frac{1}{4}\left[p^3 + \tfrac{3q^2}{p}\right]\right.$$



**Uncertain reading.** No closing square bracket is visible before the second equals sign.

Original source details: GELFAND-P000845

Kad.–Petv. eq.:

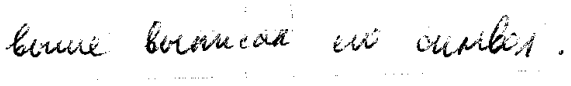
$$u_t \overset{?}{+} \frac{1}{4}\left(u_{xxx} + 6uu_x - 3\int_{-\infty}^{x} u_{yy}\,dx\right)$$



**Uncertain reading.** A vertical stroke crosses two horizontal strokes. A plus is provisional. The earlier sign is unclear.

Original source details: GELFAND-P000845

I wrote out its symbol above.



**Uncertain reading.** The Russian verb provisionally reads «выписал» (“I wrote out”). «выписан» (“is written out”) is possible.

Original source details: GELFAND-P000845

$\xi^2$ and $\xi^3$ we dress — we get the K.–P. eq..

That is, K.–P. has exactly this disp. law (as in the example)

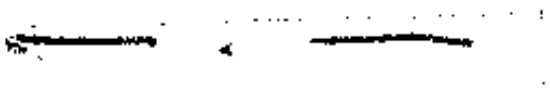


**Diagram and typed description.** —— . ——

Original source details: GELFAND-P000845

Experim. fact: we dress a pair

$$L_0 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \frac{\partial}{\partial x}, \qquad A_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \frac{\partial^2}{\partial x^2}$$

We get

$$L = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \frac{\partial}{\partial x} + \begin{bmatrix} 0 & \psi \\ \bar{\psi} & 0 \end{bmatrix}$$

and the eq.

$$\begin{cases} i\dfrac{\partial \psi}{\partial t} - \dfrac{\partial^2 \psi}{\partial x^2} + \dfrac{\partial^2 \psi}{\partial y^2} + u\psi = 0 \\ \left(\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2}\right) u = \left(\dfrac{\partial^2}{\partial x^2} - \dfrac{\partial^2}{\partial y^2}\right) |\psi|^2 \end{cases} \tag{6}$$

D...–Stewartson syst. (see hydrodyn., theory of surface waves)



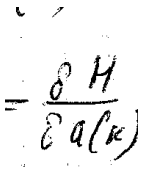
**Uncertain reading.** The first name begins with the Russian «Д» (“D”). Its continuation is indistinct.

Original source details: GELFAND-P000845

Take the Fourier trans. We get a syst. of the form:

$$\psi \to a(k)$$

$$i\frac{\partial a(k)}{\partial t} = \frac{\delta H}{\delta a(k)}$$



**Uncertain reading.** There is no visible bar over $a(k)$ in the denominator of the variational derivative.

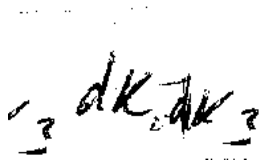
Original source details: GELFAND-P000845

$$H = \int \omega_k |a_k|^2 \, dk + \int w^{[\Gamma]}_{kk_1k_2k_3} \bar{a}_k \bar{a}_{k_1} a_{k_2} a_{k_3} \delta_{k+k_1-k_2-k_3} \, dk_2 \, dk_3$$



**Uncertain reading.** The coefficient is provisionally read $w$. A separate mark at upper right resembles $\Gamma$.

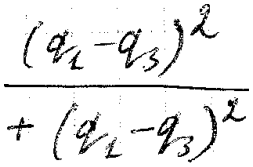
Original source details: GELFAND-P000845



**Uncertain reading.** Two differentials are visible at the frame edge. The first subscript is provisionally read as $2$.

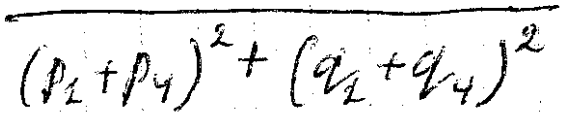
Original source details: GELFAND-P000845

$$w = \frac{(p_1 - p_3)^2 - (q_1 - q_3)^2}{(p_1 - p_3)^2 + (q_1 - q_3)^2} + \frac{(p_2 - p_4)^2 - (q_2 - q_4)^2}{(p_2 + p_4)^2 + (q_2 + q_4)^2}$$



**Uncertain reading.** The subscript on the first $q$ on both lines of the first fraction is provisionally read as $1$.

Original source details: GELFAND-P000845



**Uncertain reading.** Plus signs are visible inside both parentheses of the second denominator.

Original source details: GELFAND-P000845

GELFAND-P000846

[source number: 68]

In the space of $k_1, k_2, k_3, k_4$



**Uncertain reading.** The letter with subscript $4$ is retraced. It is provisionally set as $k_4$.

Original source details: GELFAND-P000846

consider the eq.

$$\omega(k_1) + \omega(k_2) = \omega(k_3) + \omega(k_4);$$
$$k_1 + k_2 = k_3 + k_4$$

[Crossed out: ...]



**Undeciphered passage.** A short entry and a long line below the equations are heavily crossed out. The text is illegible.

Original source details: GELFAND-P000846

It turns out that $w|_\Gamma = 0$.

Arnold's conj.: This is Poincaré's theor. on integrability. (when there is a first int. indep. of the en.)

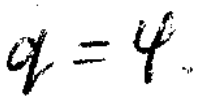
Poincaré's cond.: the ampl. of reson. harmonics must be $0$

(see Poincaré, cel. mech., or Arnold: suppl. chapters)

Poincaré: Hamiltonian

$$H = H_0(p) + \varepsilon \sum h_k(p) e^{ikq}$$

$p = I, \quad q = \psi.$



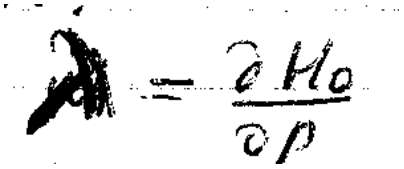
**Uncertain reading.** The angle sign is provisionally read $\psi$. $\varphi$ is possible.

Original source details: GELFAND-P000846

When $\exists$ an int. $F(p, q, \varepsilon)$ (everything is periodic in $q$),

[Under $F$:] indep. of $H$.

Nec. condition: form $\lambda = \frac{\partial H_0}{\partial p}$ and impose a nondeg. condition on it.



Original source details: GELFAND-P000846

**Uncertain reading.** The sign to the left of the derivative is heavily retraced. It is provisionally set as $\lambda$.

$$\det \left| \frac{\partial \lambda}{\partial p} \right| \neq 0$$



Original source details: GELFAND-P000846

**Uncertain reading.** The sign in the numerator is also retraced. It is provisionally set as $\lambda$.

— one of the nondeg. conditions. (there are 2 in all)

There are no integrals if

$$\Gamma: \quad \langle k, \lambda(p) \rangle = 0$$



Original source details: GELFAND-P000846

**Uncertain reading.** The sign before $(p)$ is retraced. It is provisionally set as $\lambda$.

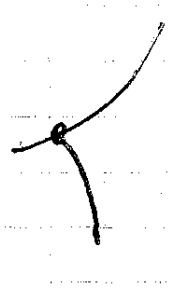
Poincaré's cond.:

$$h_k|_{\Gamma_k} = 0, \qquad k \in \mathbb{Z}^n.$$

if it is violated infinitely often, there is no integral

Idea: $\lambda \leftrightarrow \omega$.

Arn. formulas (2) mean there is a transl. surface.



Original source details: GELFAND-P000846

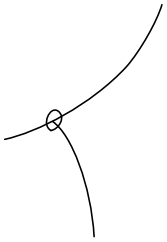


**Diagram and typed description.** The curves meet near a small loop. The thickening at left and the loop are drawn as visible. No arrows can be distinguished confidently.

we move the 2nd curve along the 1st,
translating it in parallel
Curves:

$$\begin{aligned} p &= \xi_1; & p &= -\xi_2 \\ q &= a(\xi_1); & q &= -a(\xi_2) \\ \omega &= b(\xi_1); & \omega &= -b(\xi_2) \end{aligned}$$

The curve moves by itself.
Zakharov.

$$\omega_1(k_1 + k_2) = \omega_2(k_1) + \omega_3(k_2)$$

we get general transl. surfaces.



**Diagram and typed description.**

Original source details: GELFAND-P000846

Procedure. We check whether the disp. law is degen., then see.
$0$ or not $0$ there, the Ham. interaction.



**Uncertain reading.** The abbreviation before “interaction” provisionally reads «гам.» (“Ham.”).

Original source details: GELFAND-P000846

What if the integrals start directly at higher orders?

GELFAND-P000847

Arn. Birkhoff.



**Uncertain reading.** The opening abbreviation reads «Арн.» (“Arn.”). The full stop after the following word is unclear.

Original source details: GELFAND-P000847

$$H = \sum \omega_k z_k \bar{z}_k + \dots$$

[crossed out: ~~$H =$~~]



**Uncertain reading.** $H$ can be read under the strokes. An equals sign may follow it.

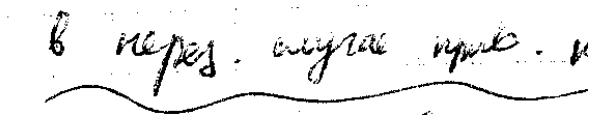
Original source details: GELFAND-P000847

$$\sum \omega_k (p_k^2 + q_k^2) + \dots$$

[At right:]

$$z = p + iq, \qquad \Pi_k = z_k \bar{z}_k$$

Birkhoff: in the nonres. case, red. to $H(\Pi_k)$.



Original source details: GELFAND-P000847

**Uncertain reading.** The abbreviation after “case” is provisionally read «прив.» (“red.”).

(if the $\omega_k$ are independent)

[At right:] (at the level of form. series)

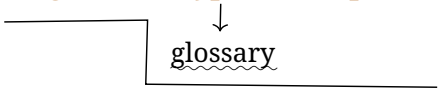
I. M. G. Short list of what one needs to know.

1. Modern abstr. language (see e.g. the 1st part of the book by Kirillov)



Original source details: GELFAND-P000847

**Diagram and typed description.**

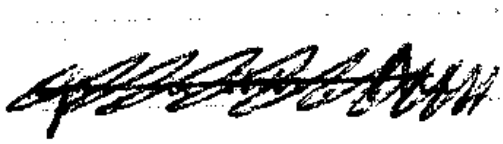



A stepped stroke and an arrow pointing to “glossary” are below the reference.

2. Arnold: diff. eqs. + suppl. chapters

(to get used to analysis)

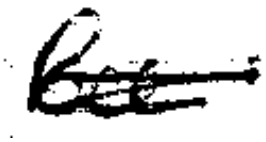
3. On algebra: 2 chapters of the book by Serre on [crossed out: ~~representations~~] reprs. of finite groups.



Original source details: GELFAND-P000847

**Uncertain reading.** The crossed-out word provisionally reads «представлениям» (“representations”).

[Above the line crossed out: ~~all~~]

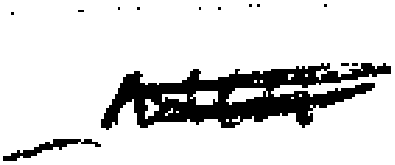


Original source details: GELFAND-P000847

**Uncertain reading.** The crossed-out word provisionally reads «все» (“all”).

all except ch. 1 of the book by Serre on Lie grps. and algebras (especially the last chapter)

[Above the line crossed out: ~~...~~]

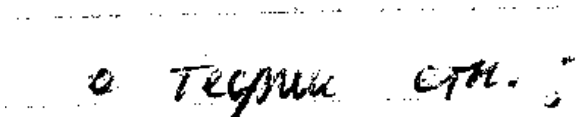


Original source details: GELFAND-P000847

**Undeciphered passage.** A short word above “Lin. algebra” is crossed out. It is illegible.

Lin. algebra: Gelfand; Maltsev

4. Diff. geometry in 2 vers.: the old tensor version (see Dirac on the theory of rel.; and Landau and Lifshitz)



Original source details: GELFAND-P000847

**Uncertain reading.** The abbreviation after "theory" provisionally reads «отн.» ("rel.").

modern vers.: vector bundles and connections.

5. Representations: H. Weyl

lectures of Raka Racah.

Gelfand, Minlos, Shapiro